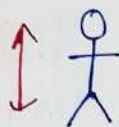


Basic Concepts (Part - I)

Introduction

Types of Vectors

Introduction :

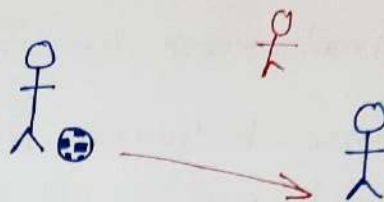
Q: What is your Height?

A: 178 cm

Magnitude only

Scalar / अदिश

e.g. length, mass, time,
distance, speed, area,
volume, temperature,
work, money, voltage,
density, resistance....



Q: How will you pass football to your teammate?

A: I will kick the football
 with sufficient muscle power
 towards my teammate.

Magnitude + Direction

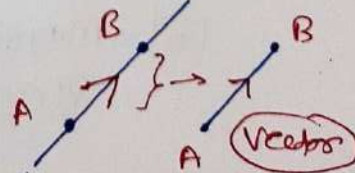
Vector / सदिश

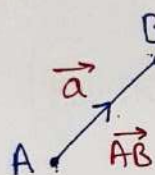
e.g. displacement, velocity, $F=mg$,
acceleration, force, weight,
momentum, electric field
intensity etc....

Line

Directed line

Directed Line Segment




 $\vec{a}$ = Directed line Segment AB = Vector AB = Vector a
 $(\vec{AB})$ $(\vec{a})$

— Arrow = Direction

— Initial point of $\vec{AB}$ = A

— Terminal point of $\vec{AB}$ = B

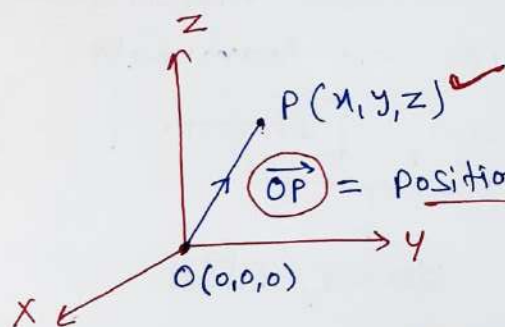
— Distance between initial and terminal points of a vector

$$\vec{a} = (\vec{AB})$$

$$|\vec{AB}| = |\vec{a}|$$

= Magnitude (length) of the vector

Position Vector of a Point (स्थिति सदिश)



$\vec{OP}$ = Position Vector of point P (with respect to O always)
Origin

$$|\vec{OP}| = \sqrt{x^2 + y^2 + z^2}$$

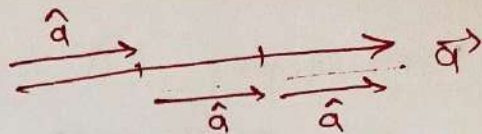
Types of Vectors →

Zero Vector (Null Vector) $(\vec{0})$ = Same initial & terminal points.

$$|\vec{0}| = \text{Magnitude} = 0$$

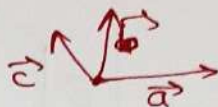
Direction = Not Defined (any direction)

Unit Vector : magnitude = 1 unit

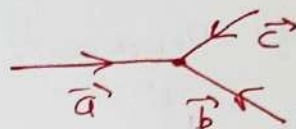


(The unit vector in the direction of $\vec{a}$ is denoted by $\hat{a}$)

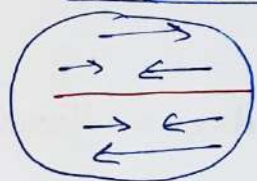
Coinitial Vectors : Two or more vectors with same initial point.



Coterminal Vectors : Two or more vectors with same terminal point.



Collinear Vectors These are parallel to the same line.



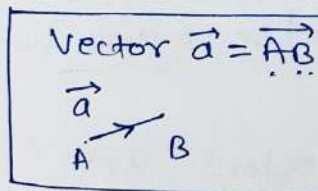
[magnitude = same or different
Direction = same or opposite]

Equal Vectors. $\vec{a} = \vec{b}$ if Magnitude = Same, direction = Same.

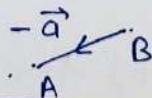


Negative of a Vector

Magnitude = Same, Direction = opposite

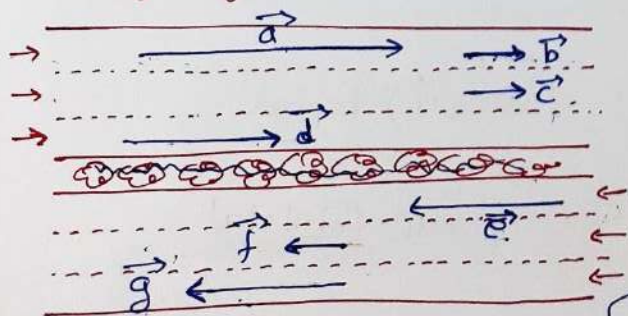


Negative vector $-\vec{a} = \overrightarrow{BA}$



$$\overrightarrow{BA} = -\overrightarrow{AB}$$

Highway



Collinear Vectors

$\vec{a}, \vec{b}, \vec{c}, \vec{d}, \vec{e}, \vec{f}, \vec{g}$

Equal Vectors

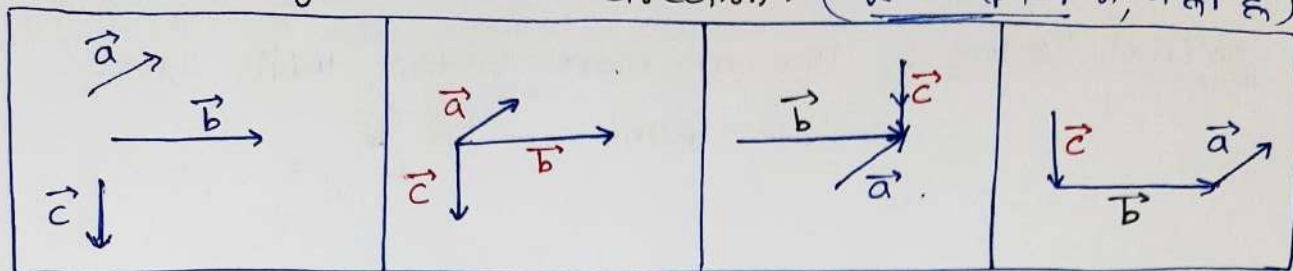
$\vec{b}, \vec{c}$ $\vec{e}, \vec{g}$

Negative of each other

$\vec{e} = -\vec{d}$

$\vec{b} = -\vec{f}$ $\vec{c} = -\vec{f}$ $\vec{g} = -\vec{d}$

Free Vectors ऐसे Vectors जिनको हम एक जगह से उठा कर दूसरी जगह shift कर सकते हैं without changing its magnitude and direction. (दूसरे chapter में, मेरी है)



Example. Represent graphically a displacement of 40 km, 20° north of west. Vector

Example. Classify the following measures as Scalar or Vector.

10 Seconds $\rightarrow$ Scalar

100 watt $\rightarrow$ Scalar

5 g/cm³ $\rightarrow$ Scalar

90° $\rightarrow$ Scalar

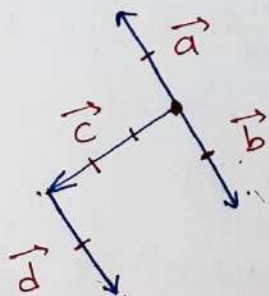
15 Newton $\rightarrow$ Vector

30 km/hr $\rightarrow$ Scalar

7 m/s² $\rightarrow$ Vector
acc.

30 km/hr towards North $\rightarrow$ Vector

Example. which of the following vectors are



a, b, d $\leftarrow$ (i) Collinear

b = d $\leftarrow$ (ii) Equal

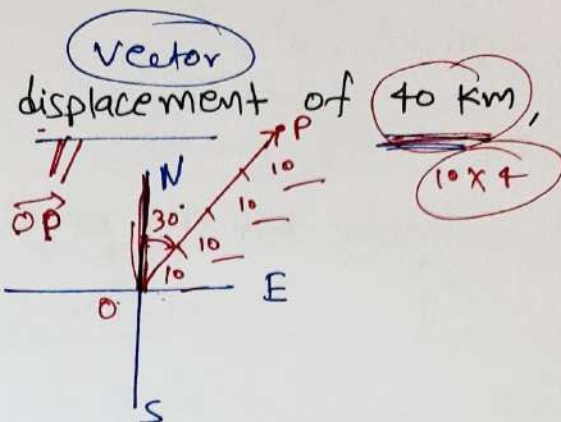
a, b, c $\leftarrow$ (iii) Coinitial

Exercise 10.1

Q.1 Represent graphically a displacement of 40 km, 30° east of north.

(length)

$$|\vec{OP}| = \text{magnitude}$$



Q.2 Classify the following as scalars and vectors.

(i) 10 kg = mass = Scalar

(ii) 2 meters north-west = Displacement = Vector

(iii) 40 = Scalar

(iv) 40 watt = power = Scalar

(v) 10^{-19} coulomb = Charge = Scalar

(vi) 20 m/s² = acceleration = Vector

Q.3 Classify the following as scalars and vectors.

(i) time period = Scalar

(ii) distance = Scalar

(iii) force = vector

(iv) velocity = vector

(v) work done = Scalar

Q.4 In Fig. (a square), identify the following vectors =

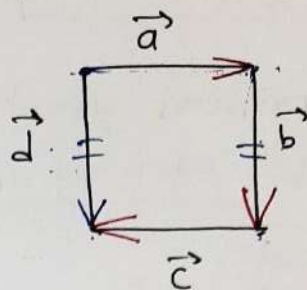
(i) coinitial $\vec{a}$ & $\vec{d}$

(ii) equal = $\vec{b} = \vec{d}$

(iii) collinear but not equal $\vec{a}, \vec{c}$

$\parallel$

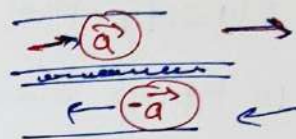
not in same direction



Q.5 Answer the following as true or false:

(i) $\vec{a}$ and $-\vec{a}$ are collinear.

$\rightarrow$ $\textcircled{T}$



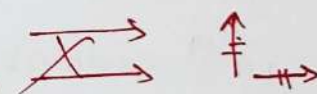
(ii) Two collinear vectors are always equal in magnitude.

$\rightarrow$ $\textcircled{F}$



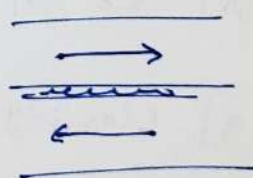
(iii) Two vectors having same magnitude are collinear.

$\rightarrow$ $\textcircled{F}$

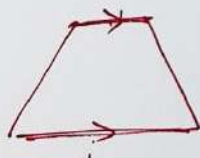


(iv) Two collinear vectors having the same magnitude are equal

$\rightarrow$ $\textcircled{F}$



Same Dirⁿ.
Same mag.

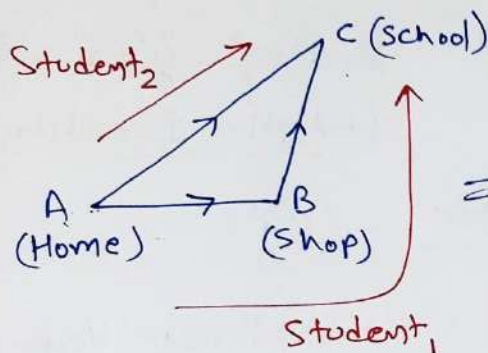


Basic Concepts (Part-II)

Distance (दूरी) = Scalar

Displacement (विस्थापन) = Vector

Addition of Vectors



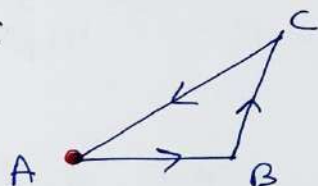
दोनों की Distance = Different

दोनों का Displacement = Same

$$\vec{AB} + \vec{BC} = \vec{AC}$$

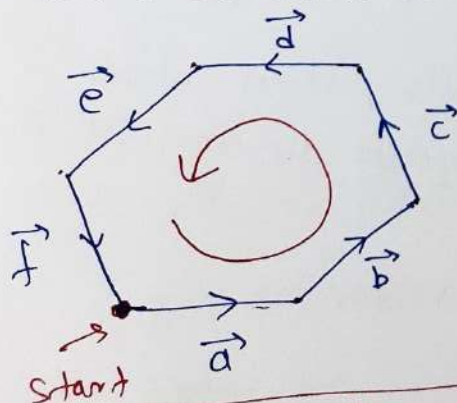
① Triangle law of Addition

Note:



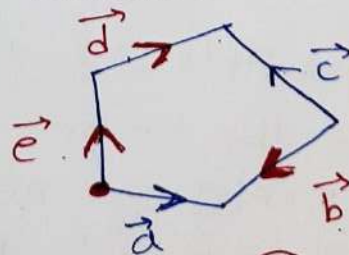
$$\vec{AB} + \vec{BC} + \vec{CA} = \vec{0} = \vec{AA}$$

Note: In a closed circuit, sum of all vectors in the same direction = $\vec{0}$



$$\vec{a} + \vec{b} + \vec{c} + \vec{d} + \vec{e} + \vec{f} = \vec{0}$$

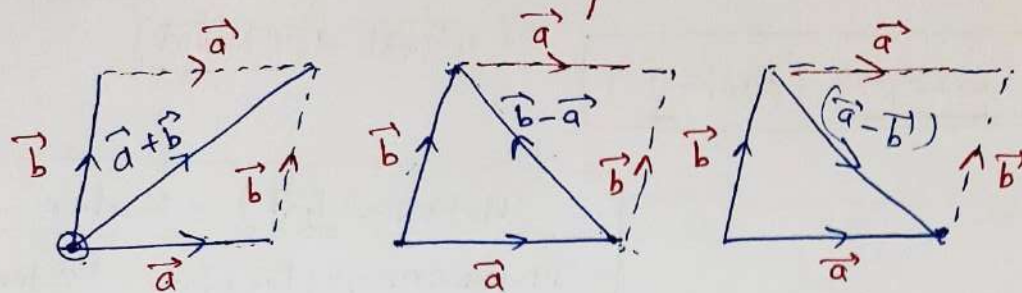
In a circuit, sum of vectors in one direction = sum of vectors in another direction



$$\vec{e} + \vec{d} + \vec{b} = \vec{a} + \vec{c}$$

$$\vec{a} + \vec{c} = \vec{e} + \vec{d} + \vec{b}$$

Parallelogram Law : $\vec{a}$ $\vec{b}$ (Free Vectors)

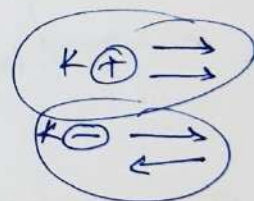
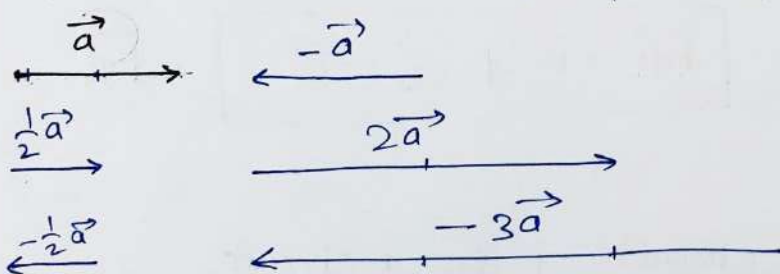


Properties of Vector Addition ① $\vec{a} + \vec{b} = \vec{b} + \vec{a}$

② $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$ ③ $\vec{a} + \vec{0} = \vec{0} + \vec{a} = \vec{a}$
 (Additive Identity)

Scalar Multiplication with Vector = $K\vec{a}$
 (Collinear Vectors)

Parallel $\vec{a} \Rightarrow \text{length } |\vec{a}|$
 $K\vec{a} \Rightarrow \text{length } |K\vec{a}| = |K||\vec{a}|$ length Different

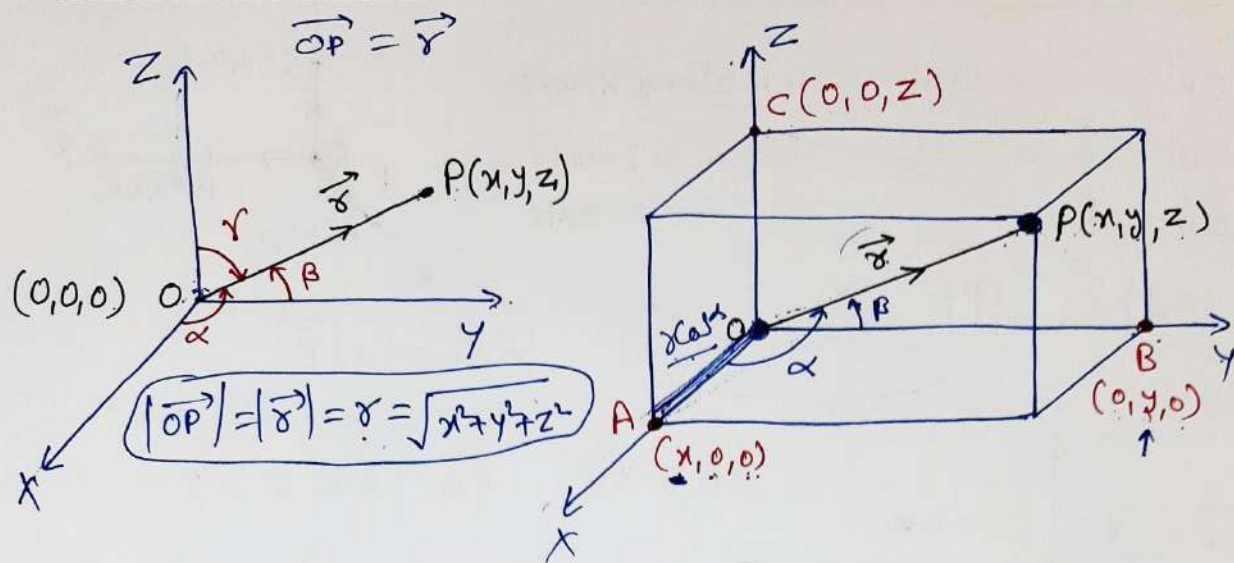


Note. Unit vector in the direction of $\vec{a}$ = $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$

e.g. If magnitude of $\vec{a}$ is 3 units, then what will be the vector in the opposite direction of $\vec{a}$ and 8 units as magnitude.

Ans. $|\vec{a}| = 3$ Required vector = $-8\frac{\vec{a}}{3}$
 $= -\frac{8}{3}\vec{a}$

Direction Cosines, Direction Ratios →



★ $\alpha, \beta, \gamma \rightarrow$ Direction angles between $\vec{r}$ and positive sides of x, y, z -axis respectively.

★ $\cos \alpha, \cos \beta, \cos \gamma \rightarrow$ Direction Cosines
 $\downarrow \quad \downarrow \quad \downarrow$
 $l \quad m \quad n$

★ $\lambda \cos \alpha, \lambda \cos \beta, \lambda \cos \gamma \rightarrow$ Direction Ratios
 $(\lambda \neq 0) (\lambda l, \lambda m, \lambda n)$ (proportional to D.C.)

By the Diagram.

$$\begin{aligned} x &= r \cos \alpha \rightarrow \cos \alpha = \frac{x}{r} = \frac{x}{\sqrt{x^2 + y^2 + z^2}} \\ y &= r \cos \beta \rightarrow \cos \beta = \frac{y}{r} = \frac{y}{\sqrt{x^2 + y^2 + z^2}} \\ z &= r \cos \gamma \rightarrow \cos \gamma = \frac{z}{r} = \frac{z}{\sqrt{x^2 + y^2 + z^2}} \end{aligned}$$

Point (x, y, z)

Note: $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

or $l^2 + m^2 + n^2 = 1$

Note: Direction Ratios → $\lambda \cos \alpha, \lambda \cos \beta, \lambda \cos \gamma$ ✓
 $\rightarrow x, y, z$ ✓
 $\rightarrow x, y, z$ ✓

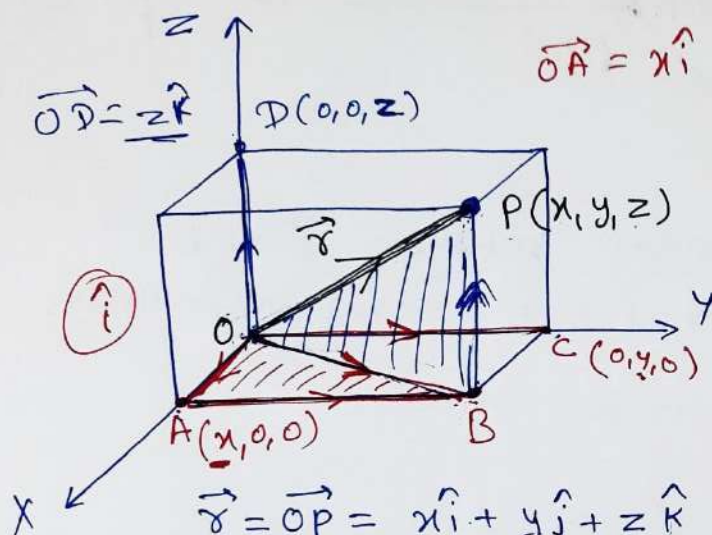
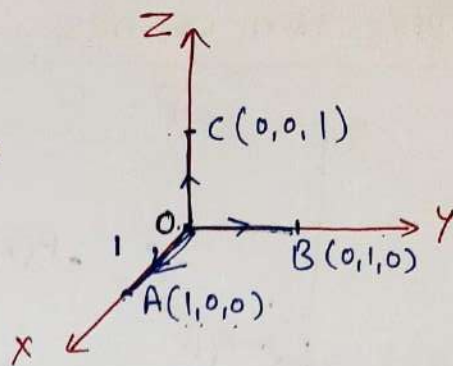
Components of a Vector $\rightarrow$

$\vec{OA} = \hat{i}$ = unit vector along x-axis

$\vec{OB} = \hat{j}$ = " " y-axis

$\vec{OC} = \hat{k}$ = " " z-axis

$$|\vec{OA}| = 1 = |\hat{i}|, |\vec{OB}| = 1 = |\hat{j}|, |\vec{OC}| = 1 = |\hat{k}|$$



$$\vec{OA} = x\hat{i}$$

$$\vec{AB} = \vec{OC} = y\hat{j}$$

$$\vec{OB} = \vec{OA} + \vec{AB} = x\hat{i} + y\hat{j}$$

$$\begin{aligned}\vec{r} &= \vec{OP} = \vec{OB} + \vec{BP} \\ &= x\hat{i} + y\hat{j} + z\hat{k}\end{aligned}$$

$$\vec{r} = \vec{OP} = x\hat{i} + y\hat{j} + z\hat{k} = \text{position vector of } P(x, y, z)$$

✓ Component Form

Vector Components
$x\hat{i}, y\hat{j}, z\hat{k}$

Scalar Components
x, y, z

$$\text{✓ length } OP = |\vec{OP}| = |\vec{r}| = |x\hat{i} + y\hat{j} + z\hat{k}| = \sqrt{x^2 + y^2 + z^2}$$

Note: $(x_1\hat{i} + y_1\hat{j} + z_1\hat{k}) + (x_2\hat{i} + y_2\hat{j} + z_2\hat{k}) = (x_1 + x_2)\hat{i} + (y_1 + y_2)\hat{j} + (z_1 + z_2)\hat{k}$

Note: Equal Vectors $\vec{a} = \vec{b}$

$$x_1\hat{i} + y_1\hat{j} + z_1\hat{k} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$$

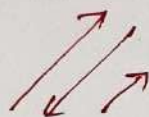
$$x_1 = x_2$$

$$y_1 = y_2$$

$$z_1 = z_2$$

Note: for collinear vectors.

$$\vec{a} = \lambda \vec{b}$$



$$\Rightarrow (x_1 \hat{i} + y_1 \hat{j} + z_1 \hat{k}) = \lambda (x_2 \hat{i} + y_2 \hat{j} + z_2 \hat{k})$$

$$x_1 = \lambda x_2$$

$$y_1 = \lambda y_2$$

$$z_1 = \lambda z_2$$

Ratio equal $\Rightarrow$

Proportional

$$\frac{x_1}{x_2} = \frac{y_1}{y_2} = \frac{z_1}{z_2} = \lambda$$

e.g. Find the unit vector in the direction of sum of the vectors, $\vec{a} = 2\hat{i} + \hat{j} - 3\hat{k}$ & $\vec{b} = \hat{i} + 4\hat{j} + 2\hat{k}$.

Ans: $\vec{a} + \vec{b} = 3\hat{i} + 5\hat{j} - \hat{k} = \vec{c}$ $|\vec{c}| = \sqrt{3^2 + 5^2 + (-1)^2}$
 $= \sqrt{9 + 25 + 1}$
 $= \sqrt{35}$

$$\hat{c} = \frac{\vec{c}}{|\vec{c}|} = \frac{3\hat{i} + 5\hat{j} - \hat{k}}{\sqrt{35}}$$

e.g. Find a vector in the direction of vector $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ that has magnitude 11 units

$$\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$$
$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{1^2 + 2^2 + 1^2}} = \frac{\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{6}}$$

Diagram showing a vector of length 11 units in the direction of $\hat{a}$. The unit vector $\hat{a}$ has length 1.

Required vector $= 11 \left(\frac{\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{6}} \right) = \frac{11}{\sqrt{6}} (\hat{i} + 2\hat{j} + \hat{k})$

e.g. Write the direction ratio's of the vector $\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$. Also find direction Cosines.

$$\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$$

Direction Ratio (DR) $= (3, -1, 2)$

$(6, -2, 4)$

$(300, -100, 200)$

D.C. $\sqrt{3^2 + (-1)^2 + 2^2} = \sqrt{14}$

$$\cos \alpha = \frac{x}{\sqrt{14}} = \frac{3}{\sqrt{14}}$$

$$\cos \beta = \frac{y}{\sqrt{14}} = \frac{-1}{\sqrt{14}}$$

$$\cos \gamma = \frac{z}{\sqrt{14}} = \frac{2}{\sqrt{14}}$$

eg. Find the values of a & b if vectors $2\hat{i} + a\hat{j} + 3\hat{k}$ and $-\hat{i} + 2\hat{j} + b\hat{k}$ are collinear.

Condⁿ. $\left(\frac{2}{-1}\right) = \frac{a}{2} = \frac{3}{b}$

$\Rightarrow -2 = \frac{a}{2} = \frac{3}{b}$

$a = -4$ ✓

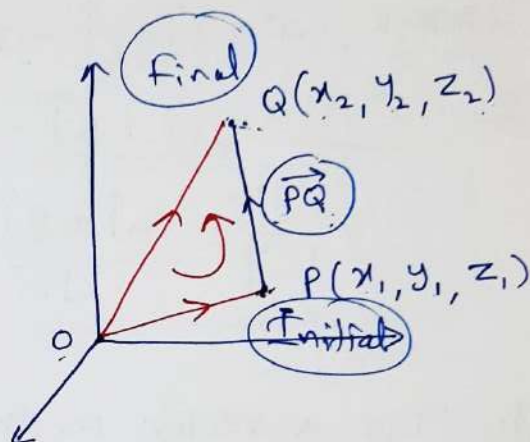
$-2 = \frac{3}{b}$

$b = \frac{3}{-2}$ ✓

* Vector Joining Two Points

$\vec{OP} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$

$\vec{OQ} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$



By Triangle law of Addition.

$\vec{OP} + \vec{PQ} = \vec{OQ}$

$\Rightarrow \boxed{\vec{PQ} = \vec{OQ} - \vec{OP}} = \text{Final} - \text{Initial}$

$\Rightarrow \vec{PQ} = (x_2\hat{i} + y_2\hat{j} + z_2\hat{k}) - (x_1\hat{i} + y_1\hat{j} + z_1\hat{k})$

$\Rightarrow \boxed{\vec{PQ} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}}$

$|\vec{PQ}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ ✓

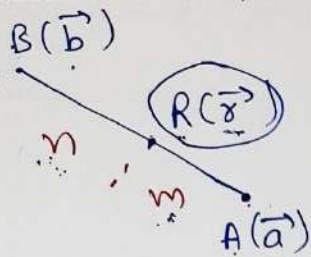
eg. Find the vector joining $A(1, 2, 3)$ and $B(3, 2, 4)$ directed from A to B.

Ans $\vec{AB} = (2)\hat{i} + (0)\hat{j} + (4-3)\hat{k}$

Final $\vec{AB} = 2\hat{i} + \hat{k}$

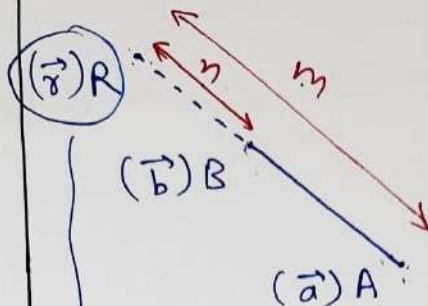
Section Formula

Internal Division



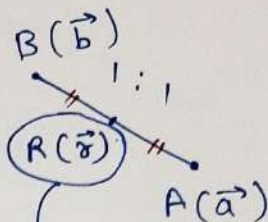
$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$$

External Division



$$\vec{r} = \frac{m\vec{b} - n\vec{a}}{m-n}$$

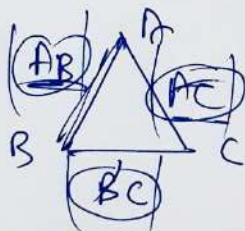
Mid Point Formula



$$\vec{r} = \frac{\vec{a} + \vec{b}}{2}$$

e.g. show that the points $A(2\hat{i} - \hat{j} + \hat{k})$, $B(\hat{i} - 3\hat{j} - 5\hat{k})$, $C(3\hat{i} - 4\hat{j} - 4\hat{k})$ are the vertices of a right angled triangle.

Final - initial



Pythagoras Theorem



$$\vec{AB} = (\hat{i} - 3\hat{j} - 5\hat{k}) - (2\hat{i} - \hat{j} + \hat{k}) = -\hat{i} - 2\hat{j} - 6\hat{k}$$

$$\vec{BC} = (3\hat{i} - 4\hat{j} - 4\hat{k}) - (\hat{i} - 3\hat{j} - 5\hat{k}) = 2\hat{i} - \hat{j} + \hat{k}$$

$$\vec{CA} = (2\hat{i} - \hat{j} + \hat{k}) - (3\hat{i} - 4\hat{j} - 4\hat{k}) = -\hat{i} + 3\hat{j} + 5\hat{k}$$

$$|\vec{AB}| = \sqrt{1+4+36} = \sqrt{41} = AB$$

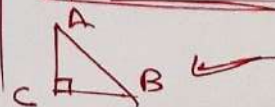
$$|\vec{BC}| = \sqrt{4+1+1} = \sqrt{6} = BC$$

$$|\vec{CA}| = \sqrt{1+9+25} = \sqrt{35} = CA$$

$$41 = 35 + 6$$

$$\Rightarrow (\sqrt{41})^2 = (\sqrt{35})^2 + (\sqrt{6})^2$$

$$\boxed{AB^2 = CA^2 + BC^2}$$



Exercise 10.2

Q.1 Compute the magnitude of the following vectors:

$$\vec{a} = \hat{i} + \hat{j} + \hat{k}, \quad \vec{b} = 2\hat{i} - 7\hat{j} - 3\hat{k}, \quad \vec{c} = \frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} - \frac{1}{\sqrt{3}}\hat{k}.$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \Rightarrow |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$|\vec{a}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}, \quad |\vec{b}| = \sqrt{(2)^2 + (-7)^2 + (-3)^2}$$

$$|\vec{c}| = \sqrt{\frac{1}{3} + \frac{1}{3} + \frac{1}{3}} = \sqrt{4 + 49 + 9}$$

$$= \sqrt{1} = 1 \quad \quad \quad = \sqrt{62}$$

Q.2 Write two ~~to~~ different vectors having same magnitude.

$$\hat{i}, \hat{j} \quad \hat{i} + \hat{j} + 2\hat{k}, \quad 2\hat{i} - \hat{j} + \hat{k}$$

Q.3 Write two different vectors having same direction.

$$\hat{i}, 2\hat{i} \quad \vec{a} \quad k\vec{a} \quad (\hat{i} + \hat{j} + 2\hat{k}), \quad (100\hat{i} + 100\hat{j} + 200\hat{k})$$

Q.4 Find the values of x & y so that vectors $2\hat{i} + 3\hat{j}$ and $x\hat{i} + y\hat{j}$ are equal.

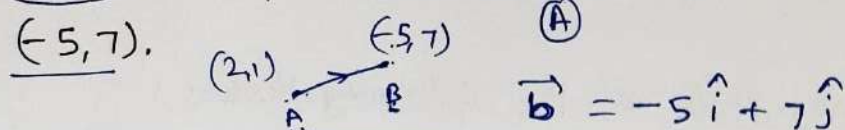
$$\vec{a} = \vec{b}$$

$$2\hat{i} + 3\hat{j} = x\hat{i} + y\hat{j}$$

$$2 = x$$

$$3 = y$$

Q.5 Find the scalar and vector components of the vector with initial point (2,1) and terminal point (-5,7).



$$\begin{aligned}\vec{AB} &= \text{Final} - \text{initial} \rightarrow \vec{a} = 2\hat{i} + \hat{j} \\ &= \vec{b} - \vec{a} = (-5\hat{i} + 7\hat{j}) - (2\hat{i} + \hat{j}) \\ &\rightarrow \vec{a} = -7\hat{i} + 6\hat{j} \\ &\text{Vector Form}\end{aligned}$$

Vector Component.
$-7\hat{i}, 6\hat{j}$
Scalar Comp.
$-7, 6$

Q.6 Find the sum of the vectors $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$,
 $\vec{b} = -2\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{c} = \hat{i} - 6\hat{j} - 7\hat{k}$.

$$\begin{aligned}\vec{a} + \vec{b} + \vec{c} &= (1-2+1)\hat{i} + (-2+4-6)\hat{j} + (1+5-7)\hat{k} \\ &= 0\hat{i} + (-4\hat{j}) + (-\hat{k}) = -4\hat{j} - \hat{k}\end{aligned}$$

Q.7 Find the unit vector in the direction of vector

$$\begin{aligned}\vec{a} &= \hat{i} + \hat{j} + 2\hat{k} \rightarrow \hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} + \hat{j} + 2\hat{k}}{\sqrt{6}} \\ |\vec{a}| &= \sqrt{1^2 + 1^2 + 2^2} = \sqrt{6} \\ &= \frac{\hat{i}}{\sqrt{6}} + \frac{\hat{j}}{\sqrt{6}} + \frac{2\hat{k}}{\sqrt{6}}\end{aligned}$$

Q.8 Find the unit vector in the direction of $\vec{PQ}$, where P & Q points are (1,2,3) and (4,5,6) respectively.

$$\text{Ans. } \vec{r} = \vec{PQ} = (\text{Final} - \text{initial}) = (4-1)\hat{i} + (5-2)\hat{j} + (6-3)\hat{k}$$

$$|\vec{r}| = |\vec{PQ}| = \sqrt{9+9+9} = \sqrt{27} = 3\sqrt{3}$$

$$\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{3\hat{i} + 3\hat{j} + 3\hat{k}}{3\sqrt{3}} = \frac{\hat{i}}{\sqrt{3}} + \frac{\hat{j}}{\sqrt{3}} + \frac{\hat{k}}{\sqrt{3}}$$

[Q.9] For given vectors, $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} - \hat{k}$, find the unit vector in the direction of the vector $\vec{a} + \vec{b}$.

Ans. $\vec{a} + \vec{b} = (2-1)\hat{i} + (-1+1)\hat{j} + (2-1)\hat{k} = \hat{i} + \hat{k} = \vec{c}$

$$\hat{c} = \frac{\vec{c}}{|\vec{c}|} = \frac{\hat{i} + \hat{k}}{\sqrt{2}} = \frac{\hat{i}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}$$

$$|\vec{c}| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

[Q.10] Find the vector in the direction of vector $5\hat{i} - \hat{j} + 2\hat{k}$ which has magnitude 8 units.

Let $\vec{a} = 5\hat{i} - \hat{j} + 2\hat{k}$

$$\hat{a} = \frac{5\hat{i} - \hat{j} + 2\hat{k}}{\sqrt{30}} = \text{unit vector in the direction of } \vec{a}$$

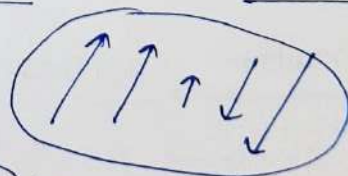
$$|\vec{a}| = \sqrt{25 + 1 + 4} = \sqrt{30}$$

vector in the direction of $\vec{a}$ & which has magnitude 8 units

$$= 8(\hat{a}) = \frac{8}{\sqrt{30}} (5\hat{i} - \hat{j} + 2\hat{k})$$

[Q.11] Show that the vectors $2\hat{i} - 3\hat{j} + 4\hat{k}$ and $-4\hat{i} + 6\hat{j} - 8\hat{k}$ are collinear.

Proportional



Ratio = equal

Condition of collinear vectors

$$\begin{pmatrix} 2 \\ -3 \\ -4 \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \\ -8 \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \\ -8 \end{pmatrix}$$

$$\Rightarrow -\frac{1}{2} = -\frac{1}{2} = -\frac{1}{2}$$

[Q.12] Find the direction Cosines of the vector $\hat{i} + 2\hat{j} + 3\hat{k}$.

(D.C.)

$$\cos \alpha = \frac{x}{\sqrt{x^2 + y^2 + z^2}} = \frac{x}{r}$$

$$\cos \beta = \frac{y}{r}$$

$$\cos \gamma = \frac{z}{r}$$

$$|\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$$

$$= \sqrt{1 + 4 + 9}$$

$$= \sqrt{14}$$



$\cos \alpha, \cos \beta, \cos \gamma$ ✓

$$\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}} \quad \checkmark$$

[Q.13] Find the direction ~~ratio~~ Cosines of the vector?

joining the points $A(1, 2, -3)$ and $B(-1, -2, 1)$, directed from A to B.



Final-initial

initial

Final

$$\vec{AB} = (-1-1)\hat{i} + (-2-2)\hat{j} + (1+3)\hat{k}$$

$$\vec{AB} = -2\hat{i} - 4\hat{j} + 4\hat{k}$$

$$|\vec{AB}| = \sqrt{x^2 + y^2 + z^2}$$

$$= \sqrt{4 + 16 + 16} = 6$$

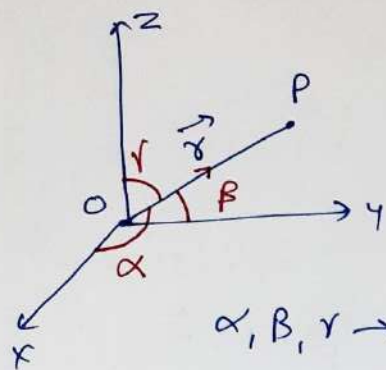
Direction Cosines.

$$\cos \alpha = \frac{x}{\sqrt{x^2 + y^2 + z^2}} = \frac{-2}{6} = -\frac{1}{3}$$

$$\cos \beta = \frac{y}{\sqrt{x^2 + y^2 + z^2}} = \frac{-4}{6} = -\frac{2}{3}$$

$$\cos \gamma = \frac{z}{\sqrt{x^2 + y^2 + z^2}} = \frac{4}{6} = \frac{2}{3}$$

Q.14 Show that the vector $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the axes OX, OY, OZ .



$$\vec{r} = \vec{OP} = \hat{i} + \hat{j} + \hat{k}$$

$$|\vec{r}| = \sqrt{1^2 + 1^2 + 1^2}$$

$$= \sqrt{3}$$

Same angle

To Prove,
 $\alpha = \beta = \gamma$

$\alpha, \beta, \gamma \rightarrow$ Direction angles between $\vec{r}$ and positive sides of x, y & z -axis.

Direction Cosines.

$$\cos \alpha = \frac{x}{\sqrt{x^2 + y^2 + z^2}} = \frac{(x)}{|\vec{r}|} = \frac{1}{\sqrt{3}}$$

$$\cos \beta = \frac{(y)}{|\vec{r}|} = \frac{1}{\sqrt{3}}$$

$$\cos \gamma = \frac{(z)}{|\vec{r}|} = \frac{1}{\sqrt{3}}$$

$$\cos \alpha = \cos \beta = \cos \gamma$$

$$\Rightarrow \alpha = \beta = \gamma$$

$$\therefore \vec{r} = \hat{i} + \hat{j} + \hat{k}$$

is equally inclined with OX, OY & OZ .

Q.15 Find the position vectors of a point R which divides the line joining two points P & Q whose position vectors are $\hat{i} + 2\hat{j} - \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ respectively in the $2:1$ ratio internally (i) externally.

(i) $2:1$

$$Q(-\hat{i} + \hat{j} + \hat{k}) = \vec{q}$$

$$P(\hat{i} + 2\hat{j} - \hat{k}) = \vec{p}$$

$$\vec{r} = \frac{2\vec{q} + \vec{p}}{2+1}$$

internal Division

$$\vec{r} = \frac{2\vec{q} + \vec{p}}{2+1}$$

$$\vec{p} = \hat{i} + 2\hat{j} - \hat{k}$$

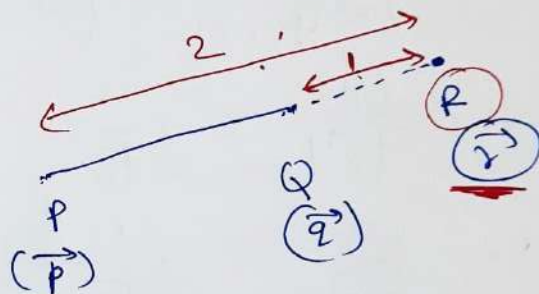
$$\vec{q} = -\hat{i} + \hat{j} + \hat{k}$$

$$\vec{r} = \frac{2(-\hat{i} + \hat{j} + \hat{k}) + (\hat{i} + 2\hat{j} - \hat{k})}{3}$$

$$\vec{r} = \frac{(-2+1)\hat{i} + (2+2)\hat{j} + (2-1)\hat{k}}{3}$$

$$\boxed{\vec{r} = -\frac{1}{3}\hat{i} + \frac{4}{3}\hat{j} + \frac{1}{3}\hat{k}}$$

(ii) external Division.



$$\vec{r} = \frac{2\vec{q} - \vec{p}}{2-1}$$

$$= \frac{2(-\hat{i} + \hat{j} + \hat{k}) - (\hat{i} + 2\hat{j} - \hat{k})}{1}$$

$$= (-2-1)\hat{i} + (2-2)\hat{j} + (2+1)\hat{k}$$

$$= -3\hat{i} + 3\hat{k}$$

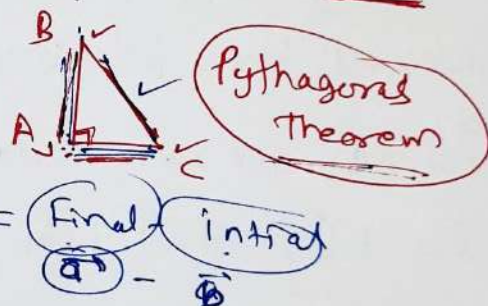
$$\underline{\underline{(-3, 0, 3)}}$$

Q.16 Find the position vectors of the mid point of the vector joining the points P(2,3,4) & Q(4,1,-2)

$\vec{P} = 2\hat{i} + 3\hat{j} + 4\hat{k}$
 $\vec{Q} = 4\hat{i} + \hat{j} - 2\hat{k}$
 $\vec{R} = \vec{OR} = \vec{r} = \frac{\vec{P} + \vec{Q}}{2} = \frac{(2\hat{i} + 3\hat{j} + 4\hat{k}) + (4\hat{i} + \hat{j} - 2\hat{k})}{2}$
 $\vec{R} = \underline{3\hat{i} + 2\hat{j} + \hat{k}}$

Q.17 Show that the ~~vectors~~ points A, B & C with position vectors, $\vec{a} = 3\hat{i} - 4\hat{j} - 4\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - 3\hat{j} - 5\hat{k}$, respectively form the vertices of a right angled triangle.

$35 = |\vec{BA}|$
 $|\vec{AC}| = \sqrt{6}$
 $|\vec{CB}| = \sqrt{41}$



$|\vec{BA}| = |\vec{a} - \vec{b}| = |3\hat{i} - 4\hat{j} - 4\hat{k} - 2\hat{i} + \hat{j} - \hat{k}|$
 $= |\hat{i} - 3\hat{j} - 5\hat{k}| = \sqrt{1^2 + 9 + 25} = \underline{(\sqrt{35})}$

$|\vec{AC}| = |\vec{c} - \vec{a}| = |\hat{i} - 3\hat{j} - 5\hat{k} - 3\hat{i} + 4\hat{j} + 4\hat{k}| = |-2\hat{i} + \hat{j} - \hat{k}|$
 $= \sqrt{4 + 1 + 1} = \underline{(\sqrt{6})}$

$|\vec{CB}| = |\vec{b} - \vec{c}| = |2\hat{i} - \hat{j} + \hat{k} - \hat{i} + 3\hat{j} + 5\hat{k}| = |\hat{i} + 2\hat{j} + 6\hat{k}|$
 $= \sqrt{1 + 4 + 36} = \underline{(\sqrt{41})}$

$41 = 35 + 6 \Rightarrow (\sqrt{41})^2 = (\sqrt{35})^2 + (\sqrt{6})^2 \Rightarrow \underline{CB^2 = BA^2 + AC^2}$

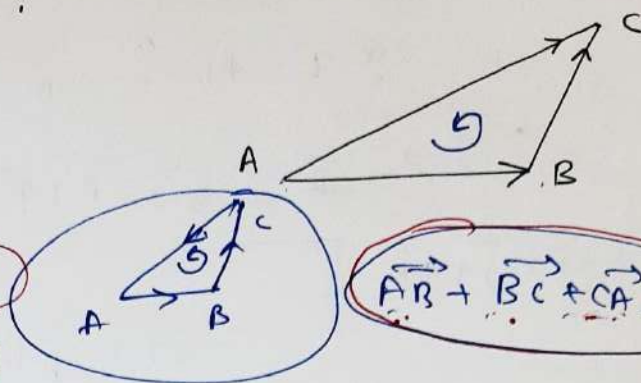
[Q.18] In triangle ABC (given fig.), which of the following is not true:

(A) $\vec{AB} + \vec{BC} + \vec{CA} = \vec{0}$ (T)

(B) $\vec{AB} + \vec{BC} - \vec{AC} = \vec{0}$ (T)

(C) $\vec{AB} + \vec{BC} - \vec{CA} = \vec{0}$ (F)

(D) $\vec{AB} - \vec{CB} + \vec{CA} = \vec{0}$ (T)



$\vec{AB} + \vec{BC} + \vec{CA} = \vec{0}$

[Q.19] If $\vec{a}$ and $\vec{b}$ are two collinear vectors, then which of the following are incorrect:

(A) $\vec{b} = \lambda \vec{a}$, for some scalar λ . (T)

(B) $\vec{a} = \pm \vec{b}$ ($|\vec{a}| = |\pm \vec{b}|$) ($|\vec{a}| = |\vec{b}|$) (F)

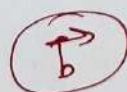
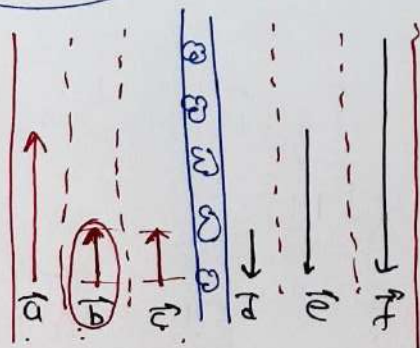
(C) the respective components of $\vec{a}$ and $\vec{b}$ are not proportional. (F)

(D) both the vectors $\vec{a}$ & $\vec{b}$ have same direction, but different magnitudes. (F)

Ans:

B, C, D

Highway



$\vec{a}, \vec{c}, \vec{d}, \vec{e}, \vec{f}$

$\lambda \vec{b}$

All vectors

$\Downarrow$

Collinear

Parallel

Product of Two Vectors

Today

Scalar (Dot) Product

$$\vec{a} \cdot \vec{b} = \text{Scalar}$$

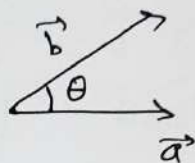
Vector (Cross) Product

$$\vec{a} \times \vec{b} = \text{Vector}$$

Scalar (or Dot) Product : (अदिश गुणनफल)

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

Scalar



Note: ① Either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, $\theta = \text{not defined}$

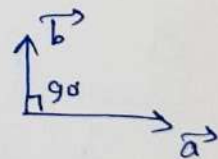
$$\vec{a} \cdot \vec{b} = 0$$

② Let $\vec{a}$ & $\vec{b}$ be non zero vectors.

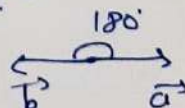
① $\theta = 0^\circ$ $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos 0^\circ = |\vec{a}| |\vec{b}|$

② $\theta = 90^\circ$ $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos 90^\circ = 0$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$



③ $\theta = 180^\circ$ $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos 180^\circ = -|\vec{a}| |\vec{b}|$



$\vec{a} \cdot \vec{a} = |\vec{a}| |\vec{a}| \cos 0^\circ = |\vec{a}|^2$

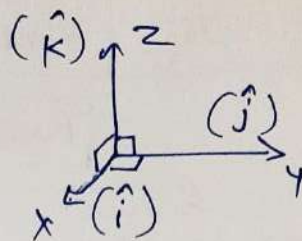
$\vec{a} \cdot \vec{a} = |\vec{a}|^2$ *

③

$$\hat{i} \cdot \hat{i} = |\hat{i}|^2 = 1^2 = 1$$

$$\hat{j} \cdot \hat{j} = 1$$

$$\hat{k} \cdot \hat{k} = 1$$



$$\hat{i} \cdot \hat{j} = 0 = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i}$$

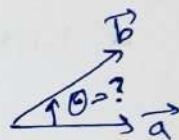
④ Commutative (क्रम विनिमय)

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

$$|\vec{a}| |\vec{b}| \cos \theta = |\vec{b}| |\vec{a}| \cos \theta$$

⑤ Dot Product $\rightarrow$ Angle

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$



$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

$$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \right)$$

$$\vec{a} \cdot (\vec{b} \pm \vec{c}) = \vec{a} \cdot \vec{b} \pm \vec{a} \cdot \vec{c}$$

$$\lambda (\vec{a} \cdot \vec{b}) = (\lambda \vec{a}) \cdot \vec{b} = \vec{a} \cdot (\lambda \vec{b})$$

Note: $(a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \cdot (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k})$
 $= a_1 b_1 + a_2 b_2 + a_3 b_3$

e.g. Find angle ' θ ' between $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.
Dot Product

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\Rightarrow (\hat{i} + \hat{j} - \hat{k}) \cdot (\hat{i} - \hat{j} + \hat{k}) = \sqrt{1^2 + 1^2 + (-1)^2} \cdot \sqrt{1^2 + 1^2 + 1^2} \cdot \cos \theta$$

$$\Rightarrow |\hat{i}|^2 - |\hat{j}|^2 - |\hat{k}|^2 = \sqrt{3} \cdot \sqrt{3} \cdot \cos \theta$$

$$\Rightarrow 1 - 1 - 1 = 3 \cdot \cos \theta$$

$$\Rightarrow \boxed{\cos \theta = -\frac{1}{3}} \quad \checkmark$$

e.g. If $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ & $\vec{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, show that

$$(\vec{a} + \vec{b}) \perp (\vec{a} - \vec{b}). \quad \vec{a} + \vec{b} = 6\hat{i} + 2\hat{j} - 8\hat{k}$$

$$\vec{a} - \vec{b} = 4\hat{i} - 4\hat{j} + 2\hat{k}$$

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = (6\hat{i} + 2\hat{j} - 8\hat{k}) \cdot (4\hat{i} - 4\hat{j} + 2\hat{k})$$

$$= 24 - 8 - 16 = 24 - 24 = 0$$

$$\boxed{(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0} \Rightarrow (\vec{a} + \vec{b}) \perp (\vec{a} - \vec{b})$$



e.g. Find $|\vec{a} - \vec{b}|$, if two vectors $\vec{a}$ & $\vec{b}$ are such that
 $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 4$. $|\vec{a} - \vec{b}| = ?$

$$|\vec{a} - \vec{b}|^2 = (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})$$

$$= \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b}$$

$$= |\vec{a}|^2 - 2(\vec{a} \cdot \vec{b}) + |\vec{b}|^2$$

$$= 4 - 2(4) + 9 = 5$$

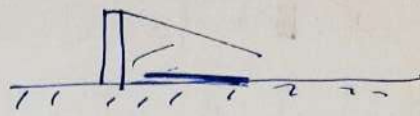
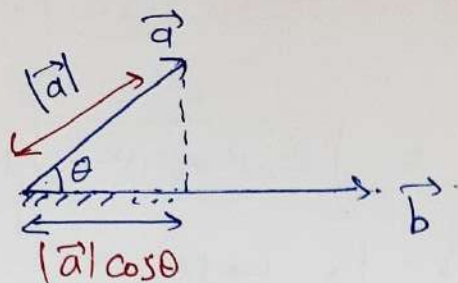
$$\boxed{|\vec{a} - \vec{b}|^2 = 5} \Rightarrow \boxed{|\vec{a} - \vec{b}| = \sqrt{5}}$$

$$\begin{array}{r} 13 \\ - 8 \\ \hline 5 \end{array}$$

$$\boxed{|\vec{a} - \vec{b}| = \sqrt{5}}$$

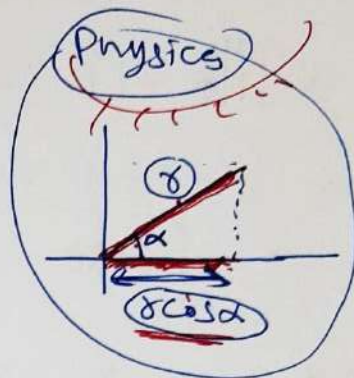
Application of Dot Product =

Projection of a Vector on Another Vector



$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\Rightarrow \boxed{\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}}$$

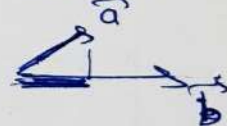


length of projection of $(\vec{a})$ on $\vec{b}$

$$= |\vec{a}| \cos \theta$$

$$= |\vec{a}| \cdot \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \right) = \frac{(\vec{a} \cdot \vec{b})}{|\vec{b}|} = \frac{\vec{a} \cdot \hat{b}}{1}$$

$$\hat{b} = \frac{\vec{b}}{|\vec{b}|}$$

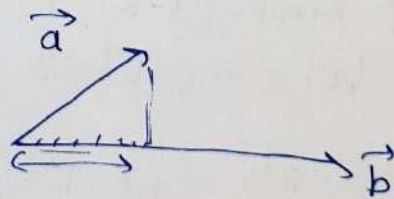


e.g. Find the projection of the vector $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on vector $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$.

Projection of $\vec{a}$ on $\vec{b}$

$$= \vec{a} \cdot \hat{b} = \vec{a} \cdot \left(\frac{\vec{b}}{|\vec{b}|} \right)$$

$$= \frac{(2\hat{i} + 3\hat{j} + 2\hat{k}) \cdot (\hat{i} + 2\hat{j} + \hat{k})}{\sqrt{1^2 + 2^2 + 1^2}} = \frac{2 + 6 + 2}{\sqrt{6}} = \frac{10}{\sqrt{6}}$$



e.g. Three vectors $\vec{a}$, $\vec{b}$ & $\vec{c}$ satisfy the condition

★ $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Evaluate the quantity.

★ $\mu = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$, if $|\vec{a}|=3$, $|\vec{b}|=4$ & $|\vec{c}|=2$.
 $= ?$

Ans. $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

$$\Rightarrow |\vec{a} + \vec{b} + \vec{c}| = |\vec{0}| = 0$$

$$\Rightarrow |\vec{a} + \vec{b} + \vec{c}|^2 = 0^2 = 0$$

$$\Rightarrow (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) = 0$$

$$\Rightarrow \underbrace{(\vec{a} \cdot \vec{a}) + \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} + \vec{c} \cdot \vec{c}}_{= 0}$$

$$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b}) + 2\vec{a} \cdot \vec{c} + 2\vec{b} \cdot \vec{c} = 0$$

$$\Rightarrow 3^2 + 4^2 + 2^2 + 2(\underbrace{\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}}_{\mu}) = 0$$

$$\Rightarrow 9 + 16 + 4 + 2(\mu) = 0$$

$$\Rightarrow 2\mu = -29 \Rightarrow \boxed{\mu = -\frac{29}{2}}$$

Exercise - 10.3

Scalar Product / (Dot Product) $\Delta\theta$

$$= \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

Q.1 Find the angle between two vectors $\vec{a}$ and $\vec{b}$ with magnitudes $\sqrt{3}$ and 2, respectively having $\vec{a} \cdot \vec{b} = \sqrt{6}$.

Ans. $\vec{a} \cdot \vec{b} = \sqrt{6}$ (given)

$$\Rightarrow |\vec{a}| |\vec{b}| \cos \theta = \sqrt{6}$$

$$\Rightarrow \sqrt{3} \cdot 2 \cos \theta = \sqrt{6} = \sqrt{2} \cdot \sqrt{3}$$

$$\Rightarrow \cos \theta = \frac{\sqrt{2} \cdot \sqrt{3}}{\sqrt{3} \cdot 2 \sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \cos \theta = \frac{1}{\sqrt{2}} \quad (\theta \in [0, \pi])$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

Q.2 Find the angle between the vectors $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} + \hat{k}$.

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\Rightarrow (\hat{i} - 2\hat{j} + 3\hat{k}) \cdot (3\hat{i} - 2\hat{j} + \hat{k}) = \sqrt{1+4+9} \cdot \sqrt{9+4+1} \cdot \cos \theta$$

$$\Rightarrow (3+4+3) = \sqrt{14} \cdot \sqrt{14} \cos \theta \Rightarrow 10 = 14 \cos \theta$$

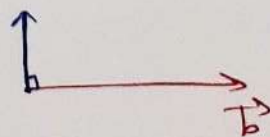
$$\Rightarrow \cos \theta = \frac{10}{14} = \left(\frac{5}{7}\right) \Rightarrow \theta = \cos^{-1}\left(\frac{5}{7}\right)$$

Q.3 Find the projection of the vector $\vec{a} = \hat{i} - \hat{j}$ on the vector $\vec{b} = \hat{i} + \hat{j}$.

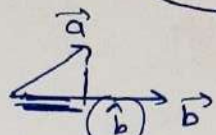
Projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

$$= \frac{(\hat{i} - \hat{j}) \cdot (\hat{i} + \hat{j})}{\sqrt{1^2 + 1^2}} = \frac{1-1}{\sqrt{2}} = \frac{0}{\sqrt{2}} = 0$$

$$\hat{b} = \frac{\vec{b}}{|\vec{b}|}$$



[Q.4] Find the projection of the vector $\hat{i} + 3\hat{j} + 7\hat{k}$ on the vector $7\hat{i} - \hat{j} + 8\hat{k}$.



Projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

$$= \frac{(\hat{i} + 3\hat{j} + 7\hat{k}) \cdot (7\hat{i} - \hat{j} + 8\hat{k})}{\sqrt{49 + 1 + 64}} = \frac{7 - 3 + 56}{\sqrt{114}} = \frac{60}{\sqrt{114}}$$

[Q.5] Show that each of the given three vectors is a unit vector:

$$\frac{1}{7}(2\hat{i} + 3\hat{j} + 6\hat{k}), \frac{1}{7}(3\hat{i} - 6\hat{j} + 2\hat{k}), \frac{1}{7}(6\hat{i} + 2\hat{j} - 3\hat{k})$$

Also, show that they are mutually perpendicular to each other.

$$|\vec{a}| = \left| \frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k} \right| = \sqrt{\frac{4}{49} + \frac{9}{49} + \frac{36}{49}} = \sqrt{\frac{49}{49}} = 1 \text{ (unit)}$$

$$|\vec{b}| = \left| \frac{3}{7}\hat{i} - \frac{6}{7}\hat{j} + \frac{2}{7}\hat{k} \right| = \sqrt{\frac{9}{49} + \frac{36}{49} + \frac{4}{49}} = \sqrt{\frac{49}{49}} = 1 \text{ (unit)}$$

$$|\vec{c}| = 1 \text{ (unit)}$$

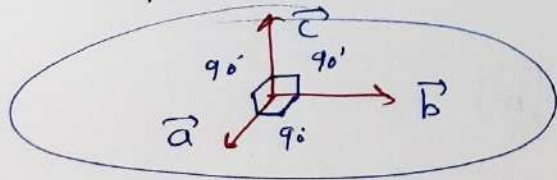
mutually Perpendicular to each other

$$\vec{a} \cdot \vec{b} = 0 \quad \vec{b} \cdot \vec{c} = 0 \quad \vec{c} \cdot \vec{a} = 0$$

$$\vec{a} \cdot \vec{b} = \frac{1}{7}(2\hat{i} + 3\hat{j} + 6\hat{k}) \cdot \frac{1}{7}(3\hat{i} - 6\hat{j} + 2\hat{k}) = \frac{1}{49}(6 - 18 + 12) = 0$$

$$\vec{b} \cdot \vec{c} = \frac{1}{7}(3\hat{i} - 6\hat{j} + 2\hat{k}) \cdot \frac{1}{7}(6\hat{i} + 2\hat{j} - 3\hat{k}) = \frac{1}{49}(18 - 12 - 6) = 0$$

$$\vec{c} \cdot \vec{a} = \frac{1}{7}(6\hat{i} + 2\hat{j} - 3\hat{k}) \cdot \frac{1}{7}(2\hat{i} + 3\hat{j} + 6\hat{k}) = \frac{1}{49}(12 + 6 - 18) = 0$$



Q.6 Find $|\vec{a}|$ and $|\vec{b}|$, if $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 8$ and $|\vec{a}| = 8|\vec{b}|$. ④

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

Ans. $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 8$

$$\Rightarrow \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} - \vec{b} \cdot \vec{b} = 8$$

$$\Rightarrow |\vec{a}|^2 - |\vec{b}|^2 = 8 \quad \text{--- ①}$$

$$|\vec{a}| = 8|\vec{b}| \quad \text{--- ②}$$

By eqn ① & ②.

$$\Rightarrow (8|\vec{b}|)^2 - |\vec{b}|^2 = 8$$

$$\Rightarrow 64|\vec{b}|^2 - |\vec{b}|^2 = 8$$

$$\Rightarrow 63|\vec{b}|^2 = 8$$

$$\Rightarrow |\vec{b}|^2 = \frac{8}{63} = \frac{4 \times 2}{9 \times 7}$$

$$\Rightarrow |\vec{b}| = \sqrt{\frac{4 \times 2}{9 \times 7}} = \frac{2\sqrt{2}}{3\sqrt{7}}$$

$$|\vec{a}| = 8|\vec{b}| = 8 \times \left(\frac{2\sqrt{2}}{3\sqrt{7}}\right)$$

$$|\vec{a}| = \frac{16\sqrt{2}}{3\sqrt{7}}$$

Q.7 Evaluate the product $(3\vec{a} - 5\vec{b}) \cdot (2\vec{a} + 7\vec{b})$.

$$\Rightarrow 6\vec{a} \cdot \vec{a} + 21\vec{a} \cdot \vec{b} - 10\vec{b} \cdot \vec{a} - 35\vec{b} \cdot \vec{b}$$

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

$$= 6|\vec{a}|^2 + 11\vec{a} \cdot \vec{b} - 35|\vec{b}|^2$$

Q.8 Find the magnitude of two vectors $\vec{a}$ and $\vec{b}$, having the same magnitude and such that the angle between them is 60° and their scalar product is $\frac{1}{2}$.

$$|\vec{a}| = |\vec{b}|$$

$$\theta = 60^\circ$$

$$\cos 60^\circ = \cos \theta = \frac{1}{2}$$

$$\vec{a} \cdot \vec{b} = \frac{1}{2}$$

$$\vec{a} \cdot \vec{b} = \frac{1}{2}$$

$$|\vec{a}| = |\vec{b}| > 0$$

$$\Rightarrow |\vec{a}| |\vec{b}| \cos \theta = \frac{1}{2}$$

$$\Rightarrow |\vec{a}| \cdot |\vec{a}| \left(\frac{1}{2}\right) = \frac{1}{2}$$

$$\Rightarrow |\vec{a}|^2 = 1 \Rightarrow |\vec{a}| = 1 = |\vec{b}|$$

[Q.9] Find $|\vec{x}|$, if for a unit vector $\vec{a}$, ~~$\vec{a}$~~

$$(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 12.$$

$$|\vec{a}| = 1$$

$$\Rightarrow \vec{x} \cdot \vec{x} + \vec{x} \cdot \vec{a} - \vec{a} \cdot \vec{x} - \vec{a} \cdot \vec{a} = 12$$

$$\vec{a} \cdot \vec{a} = |\vec{a}|^2$$

$$\Rightarrow |\vec{x}|^2 - |\vec{a}|^2 = 12$$

$$\vec{x} \cdot \vec{a} = \vec{a} \cdot \vec{x}$$

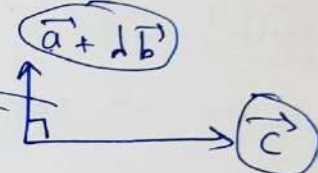
$$\Rightarrow |\vec{x}|^2 - 1^2 = 12$$

$$\Rightarrow |\vec{x}|^2 = 13 \Rightarrow |\vec{x}| = \sqrt{13}$$

$$|\vec{x}| \geq 0$$

[Q.10] If $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and

$\vec{c} = 3\hat{i} + \hat{j}$ are such that $(\vec{a} + \lambda\vec{b})$ is perpendicular to $(\vec{c})$, then find the value of λ .

$$(\vec{a} + \lambda\vec{b}) \cdot \vec{c} = 0$$


$$\Rightarrow [2\hat{i} + 2\hat{j} + 3\hat{k} + \lambda(-\hat{i} + 2\hat{j} + \hat{k})] \cdot (3\hat{i} + \hat{j}) = 0$$

$$\Rightarrow [\hat{i}(2-\lambda) + \hat{j}(2+2\lambda) + \hat{k}(3+\lambda)] \cdot (3\hat{i} + \hat{j}) = 0$$

$$\Rightarrow (2-\lambda) \cdot 3 + (2+2\lambda) \cdot 1 + 0 = 0$$

$$\Rightarrow 6 - 3\lambda + 2 + 2\lambda = 0$$

$$\Rightarrow 8 - \lambda = 0$$

$$\Rightarrow \lambda = 8$$

Q.11 Show that $|\vec{a}| \vec{b} + |\vec{b}| \vec{a}$ is perpendicular to $|\vec{a}| \vec{b} - |\vec{b}| \vec{a}$, for any two nonzero vectors $\vec{a}$ & $\vec{b}$.

$$\begin{aligned}
 & (|\vec{a}| \vec{b} + |\vec{b}| \vec{a}) \cdot (|\vec{a}| \vec{b} - |\vec{b}| \vec{a}) \\
 &= |\vec{a}|^2 \vec{b} \cdot \vec{b} - \cancel{|\vec{a}| |\vec{b}| \vec{b} \cdot \vec{a}} \\
 & \quad + \cancel{|\vec{b}| |\vec{a}| \vec{a} \cdot \vec{b}} - |\vec{b}|^2 \vec{a} \cdot \vec{a} \\
 &= \cancel{|\vec{a}|^2 |\vec{b}|^2} - \cancel{|\vec{b}|^2 |\vec{a}|^2} \\
 &= 0
 \end{aligned}$$

Diagram showing two vectors, $|\vec{a}| \vec{b} + |\vec{b}| \vec{a}$ and $|\vec{a}| \vec{b} - |\vec{b}| \vec{a}$, originating from the same point. A right-angle symbol is drawn between them, indicating they are perpendicular. A red circle contains the identity $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$.

Q.12 If $\vec{a} \cdot \vec{a} = 0$ and $\vec{a} \cdot \vec{b} = 0$, then what can be concluded about the vector $\vec{b}$?

$$\vec{a} \cdot \vec{a} = 0$$

$$\Rightarrow |\vec{a}|^2 = 0$$

$$\Rightarrow |\vec{a}| = 0$$

length of $\vec{a} = 0$

$\vec{a} = \text{Zero vector}$

magnitude $\rightarrow$ length of vector

$$\vec{a} \cdot \vec{b} = 0 \quad (\text{Given})$$

$$\Rightarrow |\vec{a}| |\vec{b}| \cos \theta = 0$$

$$\Rightarrow 0 \cdot |\vec{b}| \cos \theta = 0$$

$$0 \times () = 0$$

$\vec{b} \rightarrow \text{any vector}$

Exercise 10.3

Scalar Product (Dot Product) $\triangle \theta$

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

Q.13 If $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then find the value of $(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$.

$$|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$$

Given $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

$$|\vec{0}| = 0$$

$$|\vec{x}|^2 = \vec{x} \cdot \vec{x}$$

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

$$\Rightarrow |\vec{a} + \vec{b} + \vec{c}| = 0$$

$$\Rightarrow |\vec{a} + \vec{b} + \vec{c}|^2 = 0$$

$$\Rightarrow (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) = 0$$

$$\Rightarrow (\vec{a} \cdot \vec{a}) + \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{a} + (\vec{b} \cdot \vec{b}) + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} + (\vec{c} \cdot \vec{c}) = 0$$

$$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2\vec{a} \cdot \vec{b} + 2\vec{c} \cdot \vec{a} + 2\vec{b} \cdot \vec{c} = 0$$

$$\Rightarrow 1 + 1 + 1 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = \frac{-3}{2}$$

Q.14 If either vector $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then $\vec{a} \cdot \vec{b} = 0$.

But the Converse need not be true. Justify your answer with an example.

either $\vec{a} = \vec{0}$
or $\vec{b} = \vec{0}$

Statement

Given

If either vector $\vec{a} = \vec{0}$
or $\vec{b} = \vec{0}$

then $\vec{a} \cdot \vec{b} = 0$

$\vec{a} \cdot \vec{b} = 0$

Converse ~~Statement~~ Statement:

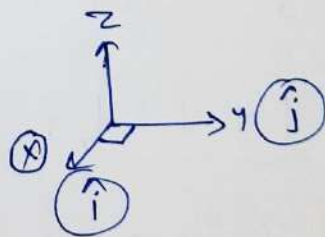
If $\vec{a} \cdot \vec{b} = 0$, then either vector $\vec{a} = \vec{0}$
or $\vec{b} = \vec{0}$

False

Let $\vec{a} = \hat{i}$
 $\vec{b} = \hat{j}$

$$\vec{a} \cdot \vec{b} = \hat{i} \cdot \hat{j} = 0$$

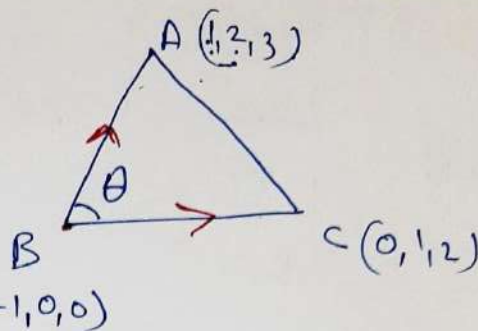
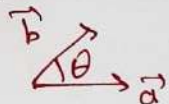
$$\vec{a} = \hat{i} \neq \vec{0}$$
$$\vec{b} = \hat{j} \neq \vec{0}$$



Q.15 If the vertices A, B, C of a triangle ABC are (1, 2, 3), (-1, 0, 0), (0, 1, 2), respectively, then find $\angle ABC$.

Dot Product

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

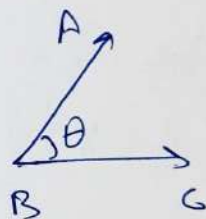


Final - initial

$$\begin{aligned} \vec{BA} &= (1+1)\hat{i} + (2-0)\hat{j} + (3-0)\hat{k} \\ &= 2\hat{i} + 2\hat{j} + 3\hat{k} \end{aligned}$$

$$\begin{aligned} \vec{BC} &= (0+1)\hat{i} + (1-0)\hat{j} + (2-0)\hat{k} \\ &= \hat{i} + \hat{j} + 2\hat{k} \end{aligned}$$

Dot Product



$$\vec{BC} \cdot \vec{BA} = |\vec{BC}| |\vec{BA}| \cos \theta$$

$$\Rightarrow (\hat{i} + \hat{j} + 2\hat{k}) \cdot (2\hat{i} + 2\hat{j} + 3\hat{k}) = \sqrt{1^2 + 1^2 + 2^2} \cdot \sqrt{4 + 4 + 9} \cdot \cos \theta$$

$$\Rightarrow (2 + 2 + 6) = \sqrt{6} \sqrt{17} \cos \theta$$

$$\Rightarrow 10 = \sqrt{102} \cdot \cos \theta$$

$$\Rightarrow \frac{10}{\sqrt{102}} = \cos \theta \Rightarrow \theta = \cos^{-1}\left(\frac{10}{\sqrt{102}}\right)$$

[Q.16] Show that the points $A(1, 2, 7)$, $B(2, 6, 3)$ and $C(3, 10, -1)$ are collinear.

Vector

Final - Initial

$$\vec{AB} = (1)\hat{i} + (4)\hat{j} + (-4)\hat{k}$$

$$\vec{BC} = (1)\hat{i} + (4)\hat{j} + (-4)\hat{k}$$

$$\vec{CA} = (-2)\hat{i} + (-8)\hat{j} + (8)\hat{k}$$

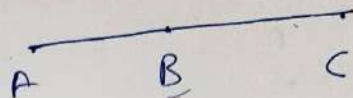
$$2\sqrt{33} = \sqrt{33} + \sqrt{33}$$

$$|\vec{CA}| = |\vec{BC}| + |\vec{AB}|$$

Collinear

points.

A, B, C



$$\vec{AB} + \vec{BC} = \vec{AC}$$

$\uparrow$ $\uparrow$ $\uparrow$
 $|\vec{AB}|$ $|\vec{BC}|$ $|\vec{AC}|$

$$\Rightarrow \vec{AB} = \hat{i} + 4\hat{j} - 4\hat{k}$$

$$|\vec{AB}| = \sqrt{1+16+16} = \sqrt{33}$$

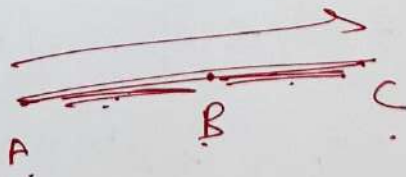
$$\vec{BC} = \hat{i} + 4\hat{j} - 4\hat{k}$$

$$|\vec{BC}| = \sqrt{1+16+16} = \sqrt{33}$$

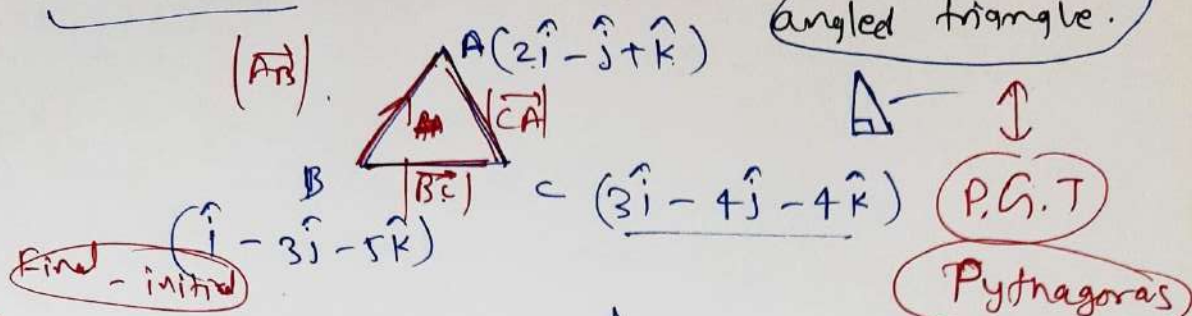
$$|\vec{CA}| = \sqrt{4+64+64}$$

$$= \sqrt{4(1+16+16)}$$

$$= 2\sqrt{33}$$



Q.17 Show that the vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$, $3\hat{i} - 4\hat{j} - 4\hat{k}$ form the vertices of a right angled triangle.



$$\vec{AB} = (\hat{i} - 3\hat{j} - 5\hat{k}) - (2\hat{i} - \hat{j} + \hat{k}) = -\hat{i} - 2\hat{j} - 6\hat{k}$$

$$\vec{BC} = (3\hat{i} - 4\hat{j} - 4\hat{k}) - (\hat{i} - 3\hat{j} - 5\hat{k}) = 2\hat{i} - \hat{j} + \hat{k}$$

$$\vec{CA} = (2\hat{i} - \hat{j} + \hat{k}) - (3\hat{i} - 4\hat{j} - 4\hat{k}) = -\hat{i} + 3\hat{j} + 5\hat{k}$$

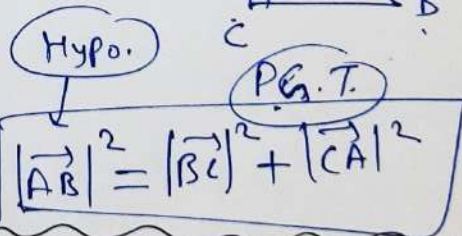
$$|\vec{AB}| = \sqrt{(-1)^2 + (-2)^2 + (-6)^2} = \sqrt{1 + 4 + 36} = \sqrt{41}$$

$$|\vec{BC}| = \sqrt{4 + 1 + 1} = \sqrt{6}$$

$$|\vec{CA}| = \sqrt{1 + 9 + 25} = \sqrt{35}$$

$$41 = 6 + 35$$

$$\Rightarrow (\sqrt{41})^2 = (\sqrt{6})^2 + (\sqrt{35})^2 \Rightarrow |\vec{AB}|^2 = |\vec{BC}|^2 + |\vec{CA}|^2$$



Q.18 If $\vec{a}$ is a non zero vector of magnitude 'a' and λ a non zero scalar, then $\lambda\vec{a}$ is unit vector

$$|\vec{a}| = a$$

$\lambda \rightarrow$ scalar $\in \mathbb{R} \setminus \{0\}$

$(\lambda\vec{a}) \rightarrow$ unit vector

- if (A) $\lambda = 1$ (B) $\lambda = -1$
(C) $a = |\lambda|$ (D) $a = 1/|\lambda|$

$$|\lambda\vec{a}| = 1$$

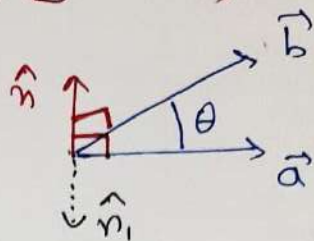
$$\Rightarrow |\lambda| |\vec{a}| = 1$$

$$\Rightarrow |\lambda| \cdot a = 1$$

$$\Rightarrow a = \frac{1}{|\lambda|}$$

Vector (or Cross) Product (सदिश गुणनफल)

$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$$



$\vec{a}, \vec{b}, \hat{n} \rightarrow$ Follow $\rightarrow$ Right Hand Thumb Rule (RHTR)
(सही order में)

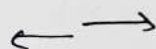
$\hat{n}$ = unit vector perpendicular to the vectors $\vec{a}$ & $\vec{b}$ obtained by RHTR

$$\vec{b} \times \vec{a} = |\vec{b}| |\vec{a}| \sin \theta (\hat{n}_1)$$

$\vec{b}, \vec{a}, \hat{n}_1 \rightarrow$ follow $\rightarrow$ Right Hand Thumb Rule.

Here $\hat{n}$ & $\hat{n}_1$ have same magnitude (1) but opposite direction.

$$\hat{n}_1 = -\hat{n}$$



$$\vec{b} \times \vec{a} = -(|\vec{b}| |\vec{a}| \sin \theta \hat{n})$$

$$\vec{b} \times \vec{a} = -(\vec{a} \times \vec{b})$$

$$\vec{a} \times \vec{b} = \hat{i} - 2\hat{j} + \hat{k}$$

$$\vec{b} \times \vec{a} = -\hat{i} + 2\hat{j} - \hat{k}$$

Cases:

$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$$

$$\boxed{\theta = 0}$$

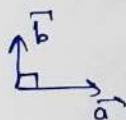
$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n} = 0$$

$$\boxed{\vec{a} \times \vec{a} = 0}$$

$$\vec{a} \times \vec{a} = 0 \quad \theta = 0$$

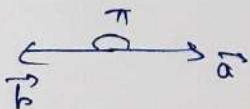
$$\vec{b} \times \vec{a} = 0 \quad \theta = 0$$

$$\theta = 90^\circ$$



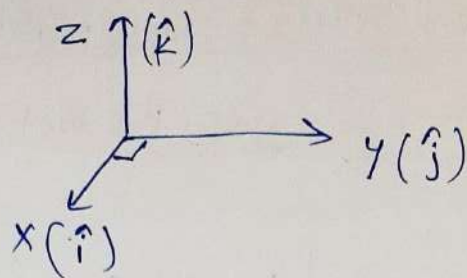
$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \hat{n}$$

$$\theta = 180^\circ$$



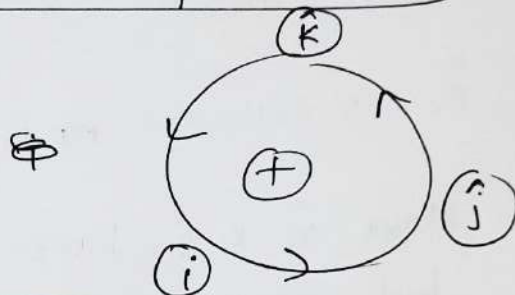
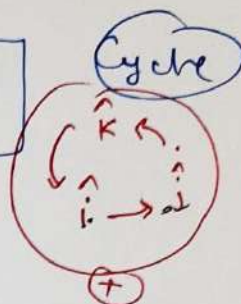
$$\vec{a} \times \vec{b} = 0$$

$$\vec{a} \times \vec{a} = 0$$



$$\begin{aligned} \hat{i} \times \hat{i} &= 0 \\ \hat{j} \times \hat{j} &= 0 \\ \hat{k} \times \hat{k} &= 0 \end{aligned}$$

$\hat{i} \times \hat{j} = \hat{k}$	$\hat{j} \times \hat{k} = \hat{i}$	$\hat{k} \times \hat{i} = \hat{j}$
$\hat{j} \times \hat{i} = -\hat{k}$	$\hat{k} \times \hat{j} = -\hat{i}$	$\hat{i} \times \hat{k} = -\hat{j}$



Note: $\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$ Cross product.

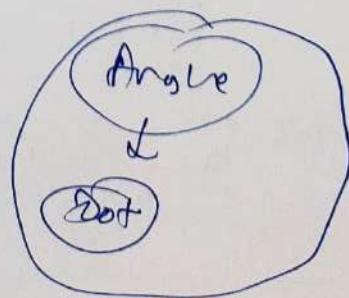
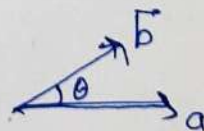
magnitude.

$$|\vec{a} \times \vec{b}| = \left| \underbrace{|\vec{a}| |\vec{b}| \sin \theta}_{\text{Scalar}} \underbrace{\hat{n}}_{\substack{\text{vector} \\ \text{Unit}}} \right|$$

$$\Rightarrow |\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta \underbrace{|\hat{n}|}_{=1}$$

$$\Rightarrow |\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

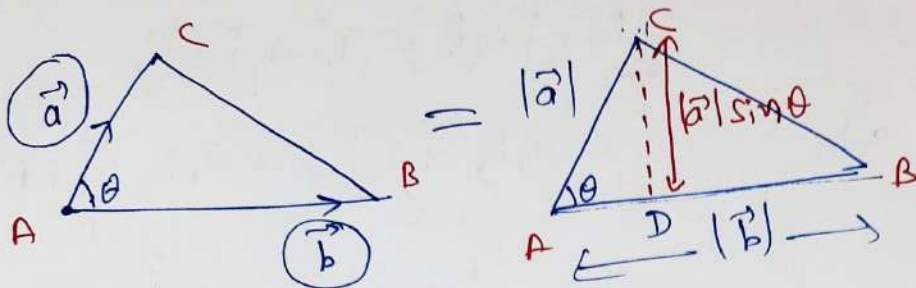
$$\Rightarrow \sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}$$



Geometric application of Vector (or Cross Product) $\Rightarrow$

$$\vec{a} \times \vec{b}$$

Area of Δ & area of parallelogram



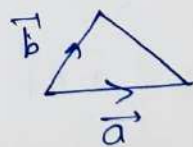
$$\text{Area of } \Delta ABC = \frac{1}{2} \text{ Base} \times \text{Height}$$

$$= \frac{1}{2} (AB) \times CD$$

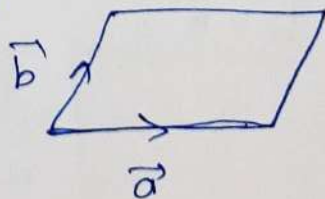
$$= \frac{1}{2} (|\vec{b}| \times |\vec{a}| \sin \theta)$$

$$= \frac{1}{2} |\vec{a} \times \vec{b}|$$

- Area of Triangle whose 2 sides are given by $\vec{a}$ & $\vec{b} = \frac{1}{2} |\vec{a} \times \vec{b}|$



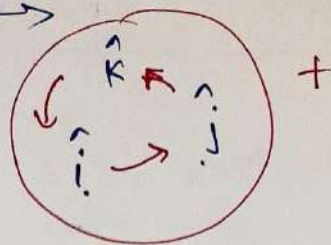
- Area of a Parallelogram whose 2 adjacent sides are given by $\vec{a}$ & $\vec{b} = |\vec{a} \times \vec{b}|$



$\vec{a} \times \vec{b}$ को solve करने के 2 तरीके $\rightarrow$

$$\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$$

$$\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$$



(i) method.

$$\vec{a} \times \vec{b} = (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \times (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k})$$

$$= a_1 b_1 (\underbrace{\hat{i} \times \hat{i}}_0) + a_1 b_2 (\underbrace{\hat{i} \times \hat{j}}_{\hat{k}}) + a_1 b_3 (\underbrace{\hat{i} \times \hat{k}}_{-\hat{j}})$$

$$+ a_2 b_1 (\underbrace{\hat{j} \times \hat{i}}_{-\hat{k}}) + 0 + a_2 b_3 (\hat{j} \times \hat{k})$$

$$+ a_3 b_1 (\hat{k} \times \hat{i}) + a_3 b_2 (\underbrace{\hat{k} \times \hat{j}}_{-\hat{i}}) + 0$$

$$\vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2) \hat{i} + (a_3 b_1 - a_1 b_3) \hat{j} + (a_1 b_2 - a_2 b_1) \hat{k}$$

II - method

$$\vec{a} \times \vec{b} = (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \times (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k})$$

$$\Rightarrow \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \xrightarrow{\text{Expand}} \text{Determinant}$$

$$\vec{b} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \end{vmatrix}$$

e.g. Find $|\vec{a} \times \vec{b}|$, if $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$ & $\vec{b} = 3\hat{i} + 5\hat{j} - 2\hat{k}$. 

Ans. $\vec{a} \times \vec{b} = (2\hat{i} + \hat{j} + 3\hat{k}) \times (3\hat{i} + 5\hat{j} - 2\hat{k})$
 $= 10\hat{k} + 4\hat{j} - 3\hat{k} - 2\hat{i} + 9\hat{j} + 15\hat{i}$

$$\vec{a} \times \vec{b} = -17\hat{i} + 13\hat{j} + 7\hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{(-17)^2 + 13^2 + 7^2}$$

$$= \sqrt{289 + 169 + 49} = \sqrt{507}$$

e.g. Find a unit vector perpendicular to each of $\star$ the $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ where $\vec{a} = \hat{i} + \hat{j} + \hat{k}$,

Ans. $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.

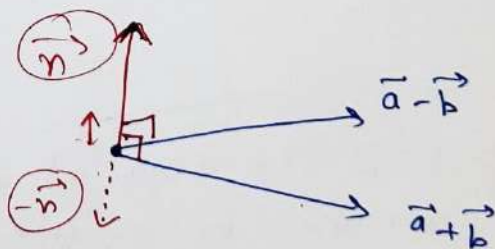
$$\vec{a} + \vec{b} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\vec{a} - \vec{b} = -\hat{j} - 2\hat{k}$$

Cross Product

vector Product

Right Hand Thumb Rule



$(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) =$ vector which is perpendicular to the both $(\vec{a} + \vec{b})$ & $(\vec{a} - \vec{b})$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 0 & -1 & -2 \end{vmatrix} = \hat{i}(-6+4) - 2\hat{i}$$

$$- \hat{j}(-4-0) = +4\hat{j}$$

$$+ \hat{k}(-2-0) - 2\hat{k}$$

$$\vec{c} = (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) = -2\hat{i} + 4\hat{j} - 2\hat{k}$$



Unit vector $\hat{c} = \frac{\vec{c}}{|\vec{c}|} = \frac{-2\hat{i} + 4\hat{j} - 2\hat{k}}{\sqrt{4 + 16 + 4}}$

$$= \frac{-2\hat{i} + 4\hat{j} - 2\hat{k}}{\sqrt{24}}$$

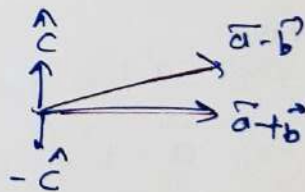
$$\hat{c} = \frac{-2\hat{i} + 4\hat{j} - 2\hat{k}}{2\sqrt{6}} = \frac{-\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{6}}$$

One more Possibility. $= -\hat{c} = \frac{\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{6}}$

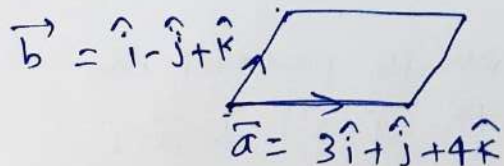
Unit vector

Perpendicular to $(\vec{a} + \vec{b})$ & $(\vec{a} - \vec{b})$

$$= \pm \left(\frac{-\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{6}} \right)$$



e.g. Find the area of a parallelogram whose sides are given by vectors $\vec{a} = 3\hat{i} + \hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$



$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 4 \\ 1 & -1 & 1 \end{vmatrix}$$

$$= \hat{i}(5) - \hat{j}(-1) + (-4)\hat{k}$$

$$\text{Required Area} = |\vec{a} \times \vec{b}|$$

$$= |5\hat{i} + \hat{j} - 4\hat{k}|$$

$$= \sqrt{25 + 1 + 16}$$

$$= \sqrt{42} \text{ Sq. units.}$$

Exercise 10.4Vector (or Cross) Product.

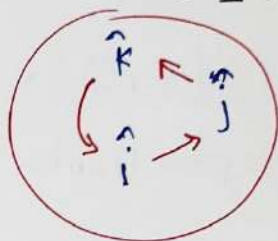
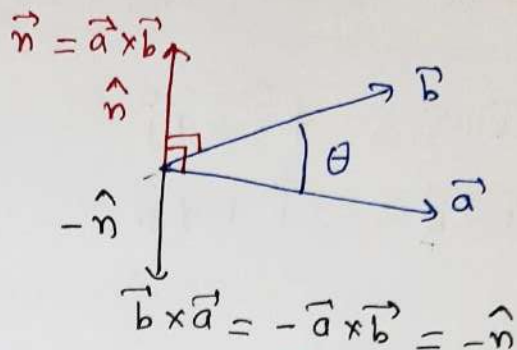
$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$$

($\vec{a}, \vec{b}, \hat{n} \rightarrow$ Right Hand thumb rule)

$$\vec{a} \times \vec{a} = 0$$

$$(\hat{i} \times \hat{i} = 0 = \hat{j} \times \hat{j} = \hat{k} \times \hat{k})$$

$$\begin{pmatrix} \hat{i} \times \hat{j} = \hat{k} \\ \hat{j} \times \hat{k} = \hat{i} \\ \hat{k} \times \hat{i} = \hat{j} \end{pmatrix} \quad \begin{pmatrix} \hat{j} \times \hat{i} = -\hat{k} \\ \hat{k} \times \hat{j} = -\hat{i} \\ \hat{i} \times \hat{k} = -\hat{j} \end{pmatrix}$$

Exercise 10.4 class 12

[Q.1] Find $|\vec{a} \times \vec{b}|$, if $\vec{a} = \hat{i} - 7\hat{j} + 7\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} + 2\hat{k}$.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -7 & 7 \\ 3 & -2 & 2 \end{vmatrix} = \hat{i} \begin{vmatrix} -7 & 7 \\ -2 & 2 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 7 \\ 3 & 2 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & -7 \\ 3 & -2 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = 0\hat{i} - \hat{j}(-19) + \hat{k}(19) = 0\hat{i} + 19\hat{j} + 19\hat{k}$$

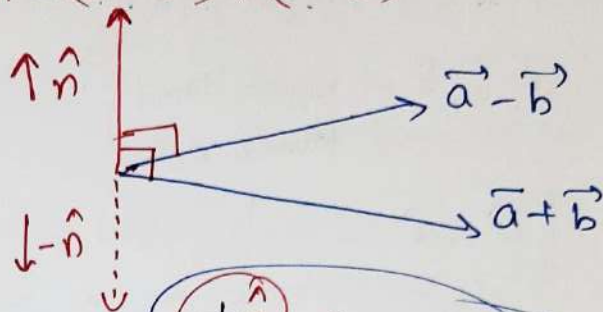
$$|\vec{a} \times \vec{b}| = \sqrt{0 + 19^2 + 19^2}$$

$$= \sqrt{2 \times 19^2} = 19\sqrt{2} \quad \checkmark$$

Q.2 Find a unit vector perpendicular to each of the vector $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$, where $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$. $\hat{n} = (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})$

$$\vec{a} + \vec{b} = 4\hat{i} + 4\hat{j}$$

$$\vec{a} - \vec{b} = 2\hat{i} + 4\hat{k}$$



$\vec{n} = (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})$ will be perpendicular to both $(\vec{a} + \vec{b})$ & $(\vec{a} - \vec{b})$.

$$\vec{n} = (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) = (4\hat{i} + 4\hat{j} + 0\hat{k}) \times (2\hat{i} + 0\hat{j} + 4\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 4 & 0 \\ 2 & 0 & 4 \end{vmatrix} = \hat{i}(16) - \hat{j}(16) + \hat{k}(-8)$$

$$\vec{n} = 16\hat{i} - 16\hat{j} - 8\hat{k}$$

$$\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{16\hat{i} - 16\hat{j} - 8\hat{k}}{\sqrt{256 + 256 + 64}} = \frac{16\hat{i} - 16\hat{j} - 8\hat{k}}{24}$$

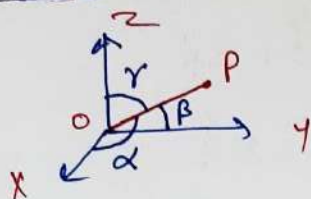
$$\hat{n} = \frac{2\hat{i}}{3} - \frac{2\hat{j}}{3} - \frac{\hat{k}}{3}$$

Perpendicular unit vectors = $\pm \hat{n} = \pm \left(\frac{2\hat{i}}{3} - \frac{2\hat{j}}{3} - \frac{\hat{k}}{3} \right)$

$$= \pm \frac{2\hat{i}}{3} \mp \frac{2\hat{j}}{3} \mp \frac{\hat{k}}{3}$$

★ Q.3 If a unit vector $\vec{a}$ makes angles $\frac{\pi}{3}$ with $\hat{i}$, $\frac{\pi}{4}$ with $\hat{j}$ and an acute angle θ with $\hat{k}$, then find θ and hence, the components of $\vec{a}$.

I-method,



$$\vec{OP} = \vec{a}$$

$\alpha, \beta, \gamma \rightarrow$ Direction angles.

$\cos \alpha, \cos \beta, \cos \gamma \rightarrow$ Direction Cosines.

$$|\vec{a}| = 1$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Unit vector $\vec{a} = \cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}$

ATQ, $\alpha = \frac{\pi}{3}, \beta = \frac{\pi}{4}, \gamma = \theta$

$$\therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\Rightarrow \cos^2 \frac{\pi}{3} + \cos^2 \frac{\pi}{4} + \cos^2 \theta = 1$$

$$\Rightarrow \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right) + \cos^2 \theta = 1$$

$$\Rightarrow \frac{1}{4} + \frac{1}{2} + \cos^2 \theta = 1$$

$$\Rightarrow \cos^2 \theta = \frac{1}{4}$$

$$\Rightarrow \cos \theta = \pm \frac{1}{2}$$

$\theta \rightarrow$ acute

$$\cos \theta = \frac{1}{2} = \cos \gamma$$

$$\theta = 60^\circ = \frac{\pi}{3}$$

$$\theta = \frac{\pi}{3}$$

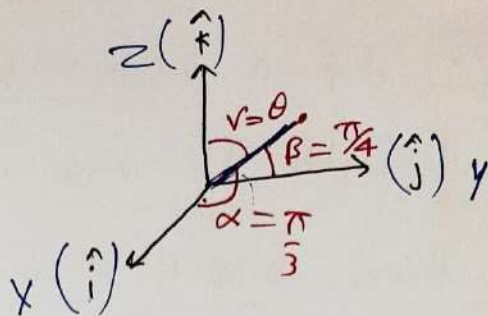
Components of $\vec{a}$

$$= \cos \alpha, \cos \beta, \cos \gamma$$

$$= \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$$

(ii) $\alpha = \pi/3$
method $\beta = \pi/4$
 $\gamma = \theta$

by Vector Product



Let unit vector $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\vec{a} \times \hat{i} = (x\hat{i} + y\hat{j} + z\hat{k}) \times \hat{i}$$

$$\Rightarrow \left| \underset{\substack{\downarrow \\ 1}}{\vec{a}} \right| \left| \underset{\substack{\downarrow \\ 1}}{\hat{i}} \right| \sin \frac{\pi}{3} \cdot \underset{\substack{\downarrow \\ 1}}{\hat{n}} = \left| -y\hat{k} + z\hat{j} \right|$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \sqrt{y^2 + z^2}$$

$$\Rightarrow \boxed{\frac{3}{4} = y^2 + z^2} \quad \text{--- (1)}$$

Similarly, $(\vec{a} \times \hat{j})$, $(\vec{a} \times \hat{k})$

$$\boxed{\frac{1}{2} = x^2 + z^2} \quad \text{--- (2)}$$

$$\boxed{\sin^2 \theta = x^2 + y^2} \quad \text{--- (3)}$$

By eqn (1) & (4): $x^2 = \frac{1}{4} \Rightarrow \boxed{x = \frac{1}{2}}$

By eqn (2) & (4): $y^2 = \frac{1}{2} \Rightarrow \boxed{y = \frac{1}{\sqrt{2}}}$

Again by eqn (4): $z^2 = \frac{1}{4} \Rightarrow \boxed{z = \frac{1}{2}}$

Components

$\therefore \vec{a} \rightarrow$ unit vector
 $(|\vec{a}| = 1)$
 $\boxed{x^2 + y^2 + z^2 = 1} \quad \text{--- (4)}$
 $\downarrow \quad \downarrow$
 $\frac{1}{4} \quad \frac{1}{2}$

$$\sin^2 \theta = x^2 + y^2$$

$$\sin^2 \theta = \frac{1}{4} + \frac{1}{2}$$

$$\sin^2 \theta = \frac{3}{4}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\boxed{\theta = \frac{\pi}{3}}$$

Q.4 Show that $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) = 2(\vec{a} \times \vec{b})$

$$\begin{aligned} \text{LHS} &= (\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) \\ &= \underbrace{\vec{a} \times \vec{a}}_{\vec{0}} + \vec{a} \times \vec{b} - \vec{b} \times \vec{a} - \underbrace{\vec{b} \times \vec{b}}_{\vec{0}} \end{aligned}$$

$$\begin{aligned} -\vec{b} \times \vec{a} \\ = \vec{a} \times \vec{b} \end{aligned}$$

$$\begin{aligned} &= (\vec{a} \times \vec{b} + \vec{a} \times \vec{b}) \\ &= 2(\vec{a} \times \vec{b}) = \text{RHS.} \end{aligned}$$

Q.5 Find λ and μ if $(2\hat{i} + 6\hat{j} + 27\hat{k}) \times (\hat{i} + \lambda\hat{j} + \mu\hat{k}) = \vec{0}$

Ans. I - method,

$$\begin{aligned} \vec{a} &\parallel k\vec{a} \\ \vec{a} \times (k\vec{a}) &= \vec{0} \end{aligned}$$

Collinear

Corresponding Components

Ratio Equal
Proportional

$$(2\hat{i} + 6\hat{j} + 27\hat{k}) \parallel (\hat{i} + \lambda\hat{j} + \mu\hat{k}) \rightarrow \text{Collinear.}$$

$$\Rightarrow \frac{2}{1} = \frac{6}{\lambda} = \frac{27}{\mu}$$

$$\frac{2}{1} = \frac{6}{\lambda}$$

$$\Rightarrow \lambda = \frac{6}{2} = 3$$

$$\frac{2}{1} = \frac{27}{\mu}$$

$$\mu = \frac{27}{2}$$

II - method

$$(2\hat{i} + 6\hat{j} + 27\hat{k}) \times (\hat{i} + \lambda\hat{j} + \mu\hat{k}) = \vec{0}$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 6 & 27 \\ 1 & \lambda & \mu \end{vmatrix} \xrightarrow{R_1} = \vec{0} = 0.\hat{i} + 0.\hat{j} + 0.\hat{k}$$

$$\Rightarrow \hat{i}(6\mu - 27\lambda) + \hat{j}(2\mu - 27) + \hat{k}(2\lambda - 6)$$
$$= 0.\hat{i} + 0.\hat{j} + 0.\hat{k}$$

Comparison

$$\hat{j}$$

$$2\mu - 27 = 0 \Rightarrow$$

$$\mu = \frac{27}{2}$$

$$\hat{k}$$

$$2\lambda - 6 = 0 \Rightarrow$$

$$\lambda = 3$$

Q.6 Given that $\vec{a} \cdot \vec{b} = 0$ & $\vec{a} \times \vec{b} = \vec{0}$. What can you conclude about the vectors $\vec{a}$ & $\vec{b}$?

$$\vec{a} \cdot \vec{b} = 0$$

$$\Rightarrow |\vec{a}| |\vec{b}| \cos \theta = 0$$

$$\theta \neq \frac{\pi}{2}$$

$$\vec{a} \times \vec{b} = \vec{0}$$

$$\Rightarrow |\vec{a}| |\vec{b}| \sin \theta \cdot \hat{n} = \vec{0}$$

$$\Rightarrow |\vec{a}| |\vec{b}| \sin \theta = 0$$

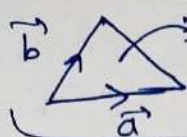
magnitude

$$\theta \neq 0$$

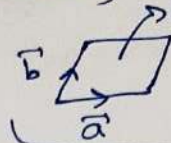
$$|\vec{a}| = 0 \text{ or } |\vec{b}| = 0$$

Exercise 10.4 → Cross Product (vector Product.)

(Q.7 to Q.12)



$$A = \frac{1}{2} |\vec{a} \times \vec{b}|$$



Q.7 Let the vectors $\vec{a}, \vec{b}, \vec{c}$ be given as $a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, $c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$. Then show that

$$\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$$

Ans. LHS = $\vec{a} \times (\vec{b} + \vec{c})$

$$= (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \times ((b_1+c_1)\hat{i} + (b_2+c_2)\hat{j} + (b_3+c_3)\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1+c_1 & b_2+c_2 & b_3+c_3 \end{vmatrix}$$

RHS = $\vec{a} \times \vec{b} + \vec{a} \times \vec{c}$

$$= (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \times (b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) + (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \times (c_1\hat{i} + c_2\hat{j} + c_3\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Property of Determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a & b \\ e & f \end{vmatrix} = \begin{vmatrix} a & b \\ c+e & d+f \end{vmatrix}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1+c_1 & b_2+c_2 & b_3+c_3 \end{vmatrix} = \text{LHS.}$$

Q.8 If either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then $\vec{a} \times \vec{b} = \vec{0}$.

Is the Converse true? ~~A~~ Justify your answer with an example.

$\theta = 0$ Collinear $\Rightarrow$

Statement

If either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$,
then $\vec{a} \times \vec{b} = \vec{0}$.

Converse Statement

If $\vec{a} \times \vec{b} = \vec{0}$
then either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$.

False,

$$\vec{a} \times \vec{b} = \vec{0}$$

either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$

Collinear vectors $\vec{a}$ & $\vec{b}$

$$\Rightarrow \theta = 0$$

$$\Rightarrow \theta = \pi$$

$$\vec{a} \times \vec{0} = \vec{0}$$

$$\vec{a} = \hat{i} \neq \vec{0}$$

$$\vec{b} = 5\hat{i} \neq \vec{0}$$

$$\begin{aligned} \vec{a} \times \vec{b} &= \vec{0} \\ \Rightarrow (\hat{i}) \times (5\hat{i}) &= \vec{0} \\ \Rightarrow 5(\vec{0}) &= \vec{0} \\ \vec{0} &= \vec{0} \end{aligned}$$

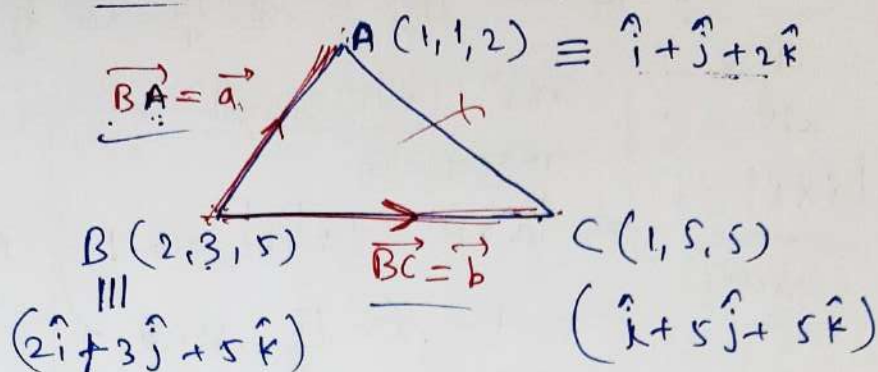
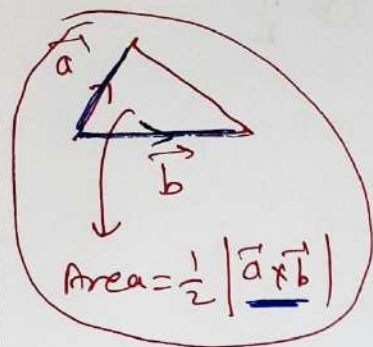
either $\vec{a} = \vec{0}$
or $\vec{b} = \vec{0}$

$$\vec{a} \times \vec{b} = \vec{0}$$

$$|\vec{a}| |\vec{b}| \sin \theta \cdot \hat{n}$$

0

[Q.9] Find the area of the triangle with vertices $A(1,1,2)$, $B(2,3,5)$ and $C(1,5,5)$.



$$\vec{a} = \vec{BA} = (-1)\hat{i} + (-2)\hat{j} + (-3)\hat{k}$$

$$\vec{a} = -\hat{i} - 2\hat{j} - 3\hat{k}$$

$$\vec{b} = \vec{BC} = -\hat{i} + 2\hat{j} + 0\hat{k}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -2 & -3 \\ -1 & 2 & 0 \end{vmatrix} = \hat{i}(6) - \hat{j}(-3) + \hat{k}(-4)$$

$$\vec{a} \times \vec{b} = 6\hat{i} + 3\hat{j} - 4\hat{k}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} |\vec{a} \times \vec{b}| = \frac{1}{2} \sqrt{36 + 9 + 16} = \frac{1}{2} \sqrt{61}$$

[Q.10] Find the area of the parallelogram whose adjacent sides are determined by the vectors $\vec{a} = \hat{i} - \hat{j} + 3\hat{k}$ & $\vec{b} = 2\hat{i} - 7\hat{j} + \hat{k}$.

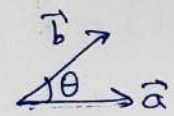
$$\vec{b} \rightarrow \text{Area} = |\vec{a} \times \vec{b}|$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 3 \\ 2 & -7 & 1 \end{vmatrix} = \hat{i}(20) - \hat{j}(-5) + \hat{k}(-5)$$

$$\text{Area} = \sqrt{400 + 25 + 25}$$

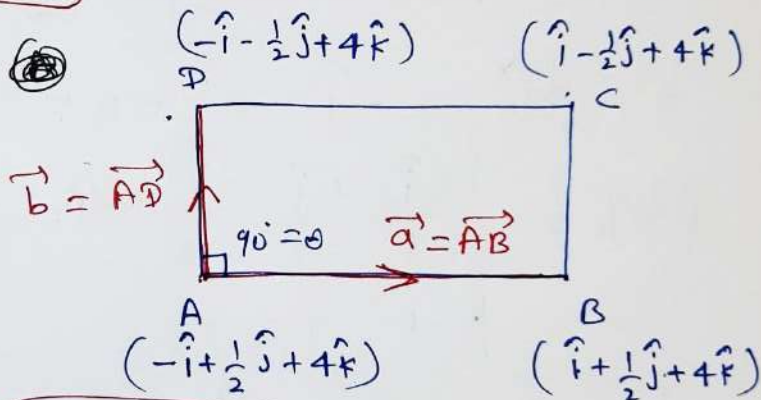
$$A = \sqrt{450} = 15\sqrt{2}$$

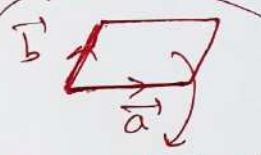
Q.11 Let the vectors $\vec{a}$ & $\vec{b}$ be such that $|\vec{a}| = 3$ & $|\vec{b}| = \frac{\sqrt{2}}{3}$, then $\vec{a} \times \vec{b}$ is a unit vector, if the angle between $\vec{a}$ & $\vec{b}$ is — (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$


 $\vec{a} \times \vec{b} = \underbrace{|\vec{a}||\vec{b}| \sin \theta}_{\text{magnitude}} \hat{n}$
 $|\vec{a} \times \vec{b}| = 1$
 (Given) $\Rightarrow |\vec{a}||\vec{b}| \sin \theta = 1$
 $\Rightarrow 1 = 3 \times \frac{\sqrt{2}}{3} \cdot \sin \theta$
 $\sin \theta = \frac{1}{\sqrt{2}}$
 $\theta = \frac{\pi}{4}$

Q.12 Area of a Rectangle having Vertices A, B, C & D with Position Vectors $-\hat{i} + \frac{\hat{j}}{2} + 4\hat{k}$, $\hat{i} + \frac{1}{2}\hat{j} + 4\hat{k}$, $\hat{i} - \frac{1}{2}\hat{j} + 4\hat{k}$, $-\hat{i} - \frac{1}{2}\hat{j} + 4\hat{k}$, respectively is —

- (A) $\frac{1}{2}$ (B) 1
(C) 2 (D) 4

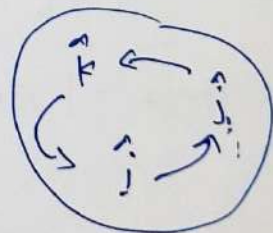



 $\text{Area} = |\vec{a} \times \vec{b}|$

Final - initial

$\vec{a} = \vec{AB} = 2\hat{i} + 0\hat{j} + 0\hat{k} = 2\hat{i}$

$\vec{b} = \vec{AD} = 0\hat{i} - \hat{j} + 0\hat{k} = -\hat{j}$



$\vec{a} \times \vec{b} = (2\hat{i}) \times (-\hat{j}) = -2(\hat{i} \times \hat{j})$

$\vec{a} \times \vec{b} = -2(\hat{k})$

$\text{Area} = |\vec{a} \times \vec{b}| = |-2\hat{k}| = \sqrt{0+0+4} = 2$

Miscellaneous Exercise on Chapter - 10

[Q.1] Write down a unit vector in XY-plane, making an angle 30° with the positive direction of x-axis.

Ans.

$$\vec{a} = \vec{OA} = ?$$

$$|\vec{a}| = 1 = |\vec{OA}| = OA$$

$$\Delta OAB \quad \cos 30^\circ = \frac{OB}{OA}$$

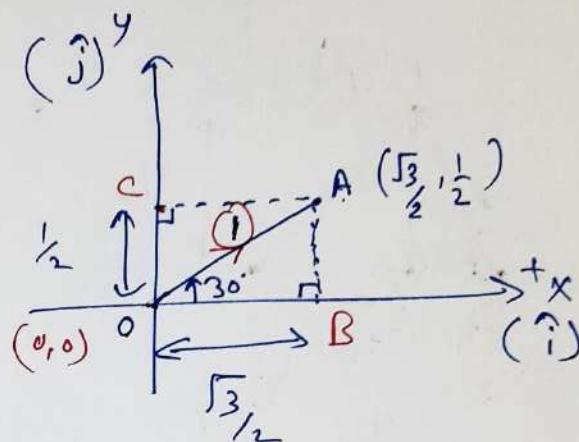
$$\Rightarrow OB = OA \cdot \cos 30^\circ$$

$$\Rightarrow OB = 1 \times \frac{\sqrt{3}}{2}$$

$$OB = \frac{\sqrt{3}}{2}$$

$$\sin 30^\circ = \frac{AB}{OA} \Rightarrow AB = \sin 30^\circ$$

$$\rightarrow 1 \Rightarrow OC = \frac{1}{2} = AB$$



$$\vec{a} = \vec{OA} = \frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}$$

[Q.2] Find the scalar components and magnitude of the vector joining the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$.

(Anal - initial)

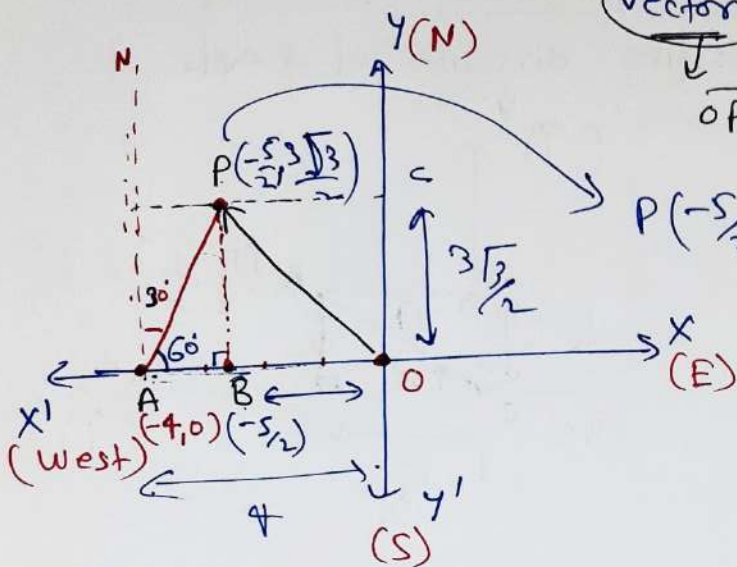
$$\vec{PQ} = (x_2 - x_1) \hat{i} + (y_2 - y_1) \hat{j} + (z_2 - z_1) \hat{k}$$

P (initial) → Q (Final)

Scalar Components = $(x_2 - x_1), (y_2 - y_1), (z_2 - z_1)$

$$|\vec{PQ}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Q-3 A girl walks 4 km towards west, then she walks 3 km in a direction 30° east of north and stops. Determine the girl's Displacement from her initial point of departure.



$$PB = \frac{3\sqrt{3}}{2}$$

$$AB = \left(\frac{3}{2} \right)$$

$$\cos 60^\circ = \frac{AB}{3} \Rightarrow AB = \frac{3}{2}$$

$$\sin 60^\circ = \frac{PB}{3} \Rightarrow PB = \frac{3\sqrt{3}}{2}$$

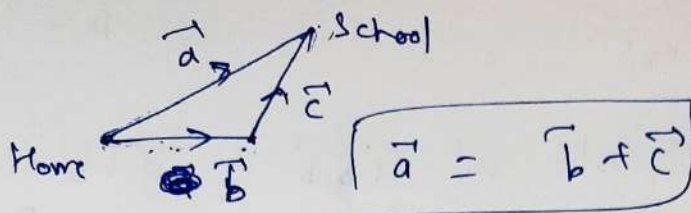
$$\begin{aligned} \text{x-coordinate of B} &= (\text{x-coordinate of A}) + \underline{AB} \\ &= (-4) + 3\frac{1}{2} = -5\frac{1}{2} \end{aligned}$$

$$\vec{OP} = -\frac{5}{2}\hat{i} + 3\sqrt{3}\frac{\hat{j}}{2}$$

$$\underline{(0,0)} \quad \underline{\left(-\frac{5}{2}, 3\frac{3}{2}\right)}$$

Q.4 If $\vec{a} = \vec{b} + \vec{c}$, then is it true that $|\vec{a}| = |\vec{b}| + |\vec{c}|$? Justify your answer.

No.



$$\vec{a} = \hat{i} + \hat{j}$$

$$\vec{b} = \hat{i}$$

$$\vec{c} = \hat{j}$$

$$|\vec{a}| = |\vec{b} + \vec{c}| \quad \checkmark$$

Here $\boxed{\vec{a} = \vec{b} + \vec{c}}$

$$\text{then } |\vec{a}| = |\vec{b}| + |\vec{c}|$$

$$|\hat{i} + \hat{j}| = |\hat{i}| + |\hat{j}|$$

$$\Rightarrow \sqrt{1+1} = 1+1$$

$$\Rightarrow \sqrt{2} = 2 \quad \text{false} \quad \checkmark$$

Q.5 Find the value of x for which $\underbrace{x(\hat{i} + \hat{j} + \hat{k})}_{\vec{a}}$ is a unit vector.

$$\vec{a} = x\hat{i} + x\hat{j} + x\hat{k}$$

$$|\vec{a}| = 1$$

$$\Rightarrow |x\hat{i} + x\hat{j} + x\hat{k}| = 1$$

$$\Rightarrow \sqrt{x^2 + x^2 + x^2} = 1$$

$$\Rightarrow 3x^2 = 1$$

$$\Rightarrow x^2 = \frac{1}{3} \Rightarrow \boxed{x = \pm \frac{1}{\sqrt{3}}}$$

Q.6 Find a vector of magnitude 5 units, and parallel to the resultant of the vectors $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ & $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$.

$$\vec{a} + \vec{b}$$

$$\vec{c} = \text{Resultant of } \vec{a} \text{ \& } \vec{b} = \vec{a} + \vec{b}$$

$$\vec{c} = (3\hat{i} + \hat{j})$$

$$|\vec{c}| = \sqrt{9+1} = \sqrt{10}$$

Here $\vec{c} = 3\hat{i} + \hat{j}$ (length = $\sqrt{10}$)

unit vector $\hat{c} = \frac{\vec{c}}{|\vec{c}|} = \left(\frac{3\hat{i} + \hat{j}}{\sqrt{10}} \right)$ (length = 1)

Required vector = $5\hat{c} = 5 \left(\frac{3\hat{i} + \hat{j}}{\sqrt{10}} \right)$ (length = 5)

$$= \cancel{5 \left(\frac{3\hat{i} + \hat{j}}{\sqrt{2} \cdot \sqrt{5}} \right)} = \frac{5(3\hat{i} + \hat{j})}{\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}}$$

$$= \cancel{\cancel{5}} \frac{(3\hat{i} + \hat{j}) \sqrt{10}}{\cancel{10}_2}$$

$$= \frac{3\sqrt{10}}{2} \hat{i} + \frac{\sqrt{10}}{2} \hat{j} \quad \checkmark$$

Q.7 If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$ &

$\vec{c} = \hat{i} - 2\hat{j} + \hat{k}$, Find a unit vector parallel to the
vector $\vec{2a} - \vec{b} + 3\vec{c}$

$$2\vec{a} - \vec{b} + 3\vec{c} = \cancel{2\hat{i}} + 2\hat{j} + 2\hat{k} - \cancel{2\hat{i}} + \hat{j} - \cancel{3\hat{k}} \\ + 3\hat{i} - 6\hat{j} + \cancel{3\hat{k}}$$

$$\vec{r} = \underline{2\vec{a} - \vec{b} + 3\vec{c}} = \underline{3\hat{i} - 3\hat{j} + 2\hat{k}} \quad \Rightarrow \quad \rightarrow$$

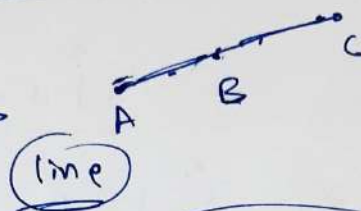
$$\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{3\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{9+9+4}} = \frac{3\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{22}}$$

Miscellaneous Exercise on Chapter -10

[Q.8] Show that the points $A(1, -2, -8)$, $B(5, 0, -2)$ and $C(11, 3, 7)$ are collinear, and find the ratio in which B divides AC.

Collinear Point

✓ (A, B, C) →



$$\underline{AB} = |\vec{AB}|$$

$$\underline{BC} = |\vec{BC}|$$

$$\underline{AC} = |\vec{AC}|$$

$$A(1, -2, -8) \quad B(5, 0, -2) \quad C(11, 3, 7)$$

$$\underline{AC} = \underline{AB} + \underline{BC}$$

Final - Initial

$$\vec{AB} = (4)\hat{i} + (2)\hat{j} + (6)\hat{k}$$

$$\vec{BC} = (6)\hat{i} + (3)\hat{j} + (9)\hat{k}$$

$$\vec{AC} = 10\hat{i} + 5\hat{j} + 15\hat{k}$$

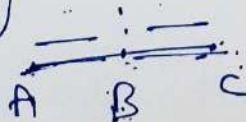
$$\rightarrow |\vec{AB}| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$|\vec{BC}| = \sqrt{36 + 9 + 81} = 3\sqrt{14}$$

Hence

$$5\sqrt{14} = 2\sqrt{14} + 3\sqrt{14}$$

$$\underline{AC} = \underline{AB} + \underline{BC}$$

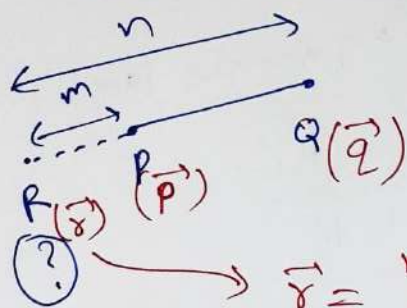


B divides AC in Ratio $\underline{AB} : \underline{BC}$

$$= 2\sqrt{14} : 3\sqrt{14}$$

$$= 2 : 3$$

Q.9 Find the position vector of a point R which divides the line joining two points P & Q whose position vectors are $(2\vec{a} + \vec{b})$ and $(\vec{a} - 3\vec{b})$ externally in the ratio 1:2. Also show that P is the mid point of the line segment RQ.



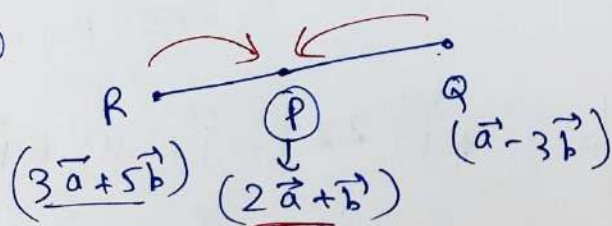
$$\vec{r} = \frac{m\vec{q} - n\vec{p}}{m - n}$$

$$\vec{r} = \frac{1 \cdot \vec{q} - 2 \cdot \vec{p}}{1 - 2} = \frac{(\vec{a} - 3\vec{b}) - 2(2\vec{a} + \vec{b})}{-1}$$

$$\vec{r} = \frac{\vec{a} - 3\vec{b} - 4\vec{a} - 2\vec{b}}{-1} = \frac{-3\vec{a} - 5\vec{b}}{-1}$$

$$\boxed{\vec{r} = 3\vec{a} + 5\vec{b}} \leftarrow \text{Position vector of 'R'}$$

Also



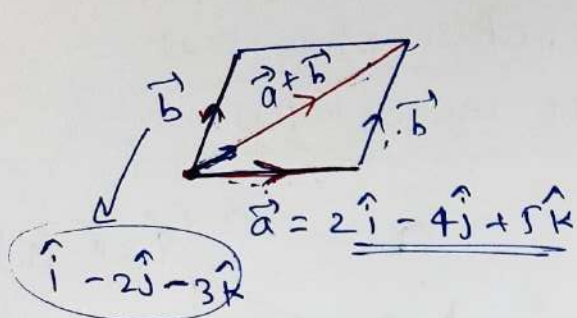
$$\text{Mid Point of R \& Q} = \frac{\vec{r} + \vec{q}}{2} = \frac{(3\vec{a} + 5\vec{b}) + (\vec{a} - 3\vec{b})}{2}$$

~~Mid Point~~

$$= 2\vec{a} + \vec{b}$$

$$= (\vec{p})$$

[Q.10] The two adjacent sides of a parallelogram are $2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\hat{i} - 2\hat{j} - 3\hat{k}$. Find the unit vector parallel to its diagonal. Also, find its Area.

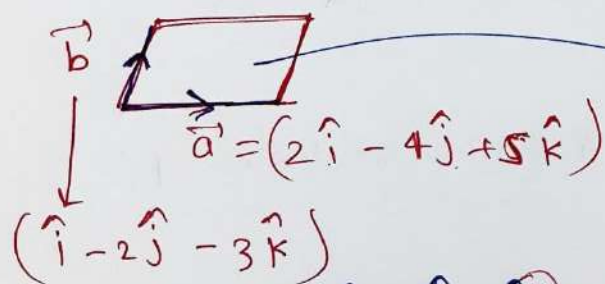


$\vec{d} = \text{Diagonal} = \vec{a} + \vec{b}$

$$= (2\hat{i} - 4\hat{j} + 5\hat{k}) + (\hat{i} - 2\hat{j} - 3\hat{k})$$

$$\vec{d} = 3\hat{i} - 6\hat{j} + 2\hat{k}$$

$$\hat{d} = \frac{\vec{d}}{|\vec{d}|} = \frac{3\hat{i} - 6\hat{j} + 2\hat{k}}{\sqrt{9 + 36 + 4}} = \left(\frac{3\hat{i} - 6\hat{j} + 2\hat{k}}{7} \right)$$



$$\text{Area} = |\vec{a} \times \vec{b}|$$

Cross Product

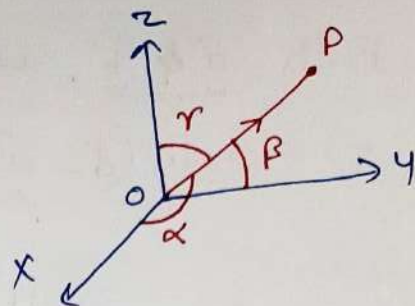
$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -4 & 5 \\ 1 & -2 & -3 \end{vmatrix} = +\hat{i}(2 \cdot 2) - \hat{j}(-11) + \hat{k}(\ominus)$$

$$= (22\hat{i} + 11\hat{j} + \ominus\hat{k})$$

$$\text{Area} = |\vec{a} \times \vec{b}| = \sqrt{484 + 121 + \ominus} = \sqrt{484 + 121 + 0}$$

$$= \sqrt{605} = \sqrt{5 \times 121} = 11\sqrt{5}$$

Q.11 Show that the direction cosines of a vector equally inclined to the axes OX, OY, OZ are $\pm \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$.



$\alpha, \beta, \gamma \rightarrow$ Direction Angles

$\cos \alpha, \cos \beta, \cos \gamma \rightarrow$ Direction Cosines Property,

$$\boxed{\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1}$$

Equally inclined $\rightarrow$ (angles $\rightarrow$ equal,)
(OX, OY, OZ)

$$\alpha = \beta = \gamma$$

$$\Rightarrow \boxed{\cos \alpha = \cos \beta = \cos \gamma} \text{ A.T.Q.}$$

$$\therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\Rightarrow \cos^2 \alpha + \cos^2 \alpha + \cos^2 \alpha = 1 \Rightarrow 3 \cos^2 \alpha = 1$$

$$\Rightarrow \cos^2 \alpha = \frac{1}{3} \Rightarrow \cos \alpha = \left(\pm \frac{1}{\sqrt{3}} \right) = \cos \beta = \cos \gamma$$

D.C. $\rightarrow \cos \alpha, \cos \beta, \cos \gamma$

$$= \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$$

Q.12 Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ &

$\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. Find a vector $\vec{d}$ which is perpendicular to both $\vec{a}$ & $\vec{b}$, and $\vec{c} \cdot \vec{d} = 15$.

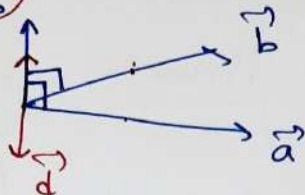
Ans. I-method $\vec{d} = x\hat{i} + y\hat{j} + z\hat{k}$ $\xrightarrow{\text{Cond}^n}$ $\begin{cases} \vec{a} \cdot \vec{d} = 0 & \text{--- (1)} \\ \vec{b} \cdot \vec{d} = 0 & \text{--- (2)} \\ \vec{c} \cdot \vec{d} = 15 & \text{--- (3)} \end{cases}$

II-method

$$(\vec{a} \times \vec{b}) \perp \vec{a}$$

$$(\vec{a} \times \vec{b}) \perp \vec{b}$$

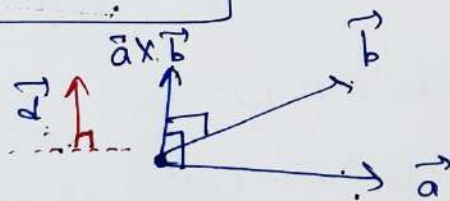
$$\vec{a} \times \vec{b}$$



$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 2 \\ 3 & -2 & 7 \end{vmatrix} = \hat{i}(32) - \hat{j}(1) + \hat{k}(-14)$$

$$= 32\hat{i} - \hat{j} - 14\hat{k}$$

$$\vec{a} \times \vec{b} = 32\hat{i} - \hat{j} - 14\hat{k}$$



Here $\vec{d}$ & $(\vec{a} \times \vec{b})$

are collinear vectors.

$$\vec{d} = \lambda(\vec{a} \times \vec{b})$$

$$\vec{d} = \lambda(32\hat{i} - \hat{j} - 14\hat{k})$$

Also $\vec{c} \cdot \vec{d} = 15$

$$\Rightarrow (2\hat{i} - \hat{j} + 4\hat{k}) \cdot \lambda(32\hat{i} - \hat{j} - 14\hat{k}) = 15$$

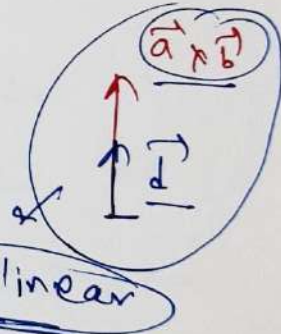
$$\Rightarrow \lambda(64 + 1 - 56) = 15$$

$$\Rightarrow \lambda(9) = 15 \Rightarrow \lambda = \frac{15}{9} = \frac{5}{3}$$

Answer

$$\vec{d} = \frac{5}{3}(32\hat{i} - \hat{j} - 14\hat{k})$$

$$\vec{d} = \frac{1}{3}(160\hat{i} - 5\hat{j} - 70\hat{k})$$



Miscellaneous Exercise on Chapter - (10)

Q.13 The scalar product of the vector $\hat{i} + \hat{j} + \hat{k}$ with a unit vector along the sum of vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\lambda\hat{i} + 2\hat{j} + 3\hat{k}$, is equal to one. Find the value of λ .

Ans.

Sum of vectors $(2\hat{i} + 4\hat{j} - 5\hat{k})$ & $(\lambda\hat{i} + 2\hat{j} + 3\hat{k})$

$$\vec{C} = (2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}$$

$$\hat{C} = \frac{\vec{C}}{|\vec{C}|} = \frac{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{(2+\lambda)^2 + 36 + 4}} = \frac{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{4 + \lambda^2 + 4\lambda + 40}}$$

$$\hat{C} = \left(\frac{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{\lambda^2 + 4\lambda + 44}} \right)$$

ATQ. Scalar Product of $(\hat{i} + \hat{j} + \hat{k})$ with $\hat{C} = 1$

$$\Rightarrow (\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{C}) = 1$$

$$\Rightarrow (\hat{i} + \hat{j} + \hat{k}) \cdot \left(\frac{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{\lambda^2 + 4\lambda + 44}} \right) = 1$$

$$\Rightarrow (2+\lambda) + 6 - 2 = \sqrt{\lambda^2 + 4\lambda + 44}$$

$$\Rightarrow (\lambda + 6)^2 = (\sqrt{\lambda^2 + 4\lambda + 44})^2$$

$$\Rightarrow \cancel{\lambda^2} + 12\lambda + 36 = \cancel{\lambda^2} + 4\lambda + 44$$

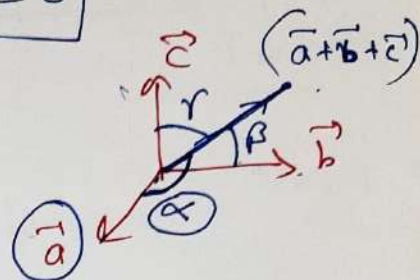
$$\Rightarrow 8\lambda = 8$$

$$\Rightarrow \lambda = 1$$

Q.14 If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors of equal magnitudes, show that the vector $(\vec{a} + \vec{b} + \vec{c})$ is equally inclined to $\vec{a}, \vec{b}$ & $\vec{c}$.

Ans Given: $\vec{a} \cdot \vec{b} = 0$, $\vec{b} \cdot \vec{c} = 0$, $\vec{a} \cdot \vec{c} = 0$
 $|\vec{a}| = |\vec{b}| = |\vec{c}| = K$ (let)

Let α, β & γ be the angles between $(\vec{a} + \vec{b} + \vec{c})$ and $\vec{a}, \vec{b}, \vec{c}$ respectively.



To Prove: (Equally inclined) $\rightarrow$ angle same

Proof:

$\alpha = \beta = \gamma$

$$(\vec{a} + \vec{b} + \vec{c}) \cdot \vec{a} = |\vec{a} + \vec{b} + \vec{c}| \cdot |\vec{a}| \cos \alpha$$

$$\Rightarrow \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a} = |\vec{a} + \vec{b} + \vec{c}| \cdot |\vec{a}| \cdot \cos \alpha$$

$$\Rightarrow |\vec{a}|^2 = |\vec{a} + \vec{b} + \vec{c}| \cdot |\vec{a}| \cdot \cos \alpha$$

$$\Rightarrow |\vec{a}| = |\vec{a} + \vec{b} + \vec{c}| \cdot \cos \alpha$$

$$\Rightarrow \cos \alpha = \frac{|\vec{a}|}{|\vec{a} + \vec{b} + \vec{c}|}$$

$$\Rightarrow \cos \alpha = \frac{K}{|\vec{a} + \vec{b} + \vec{c}|} \quad \text{--- (1)}$$

Similarly,

$$(\vec{a} + \vec{b} + \vec{c}) \cdot \vec{b}$$

$$\cos \beta = \frac{K}{|\vec{a} + \vec{b} + \vec{c}|} \quad \text{--- (2)}$$

$$\cos \gamma = \frac{K}{|\vec{a} + \vec{b} + \vec{c}|} \quad \text{--- (3)}$$

By eqn (1), (2) & (3): $\rightarrow$

$$\cos \alpha = \cos \beta = \cos \gamma \Rightarrow \alpha = \beta = \gamma$$

Q.15 Prove that $(\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = |\vec{a}|^2 + |\vec{b}|^2$, if and only if $\vec{a}, \vec{b}$ are perpendicular, given $\vec{a} \neq \vec{0}, \vec{b} \neq \vec{0}$.

Ans. $\frac{(\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b})}{= |\vec{a}|^2 + |\vec{b}|^2} \iff \underline{\vec{a} \perp \vec{b}}$

Part (I) Let $(\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = |\vec{a}|^2 + |\vec{b}|^2$

$$\Rightarrow (\vec{a} \cdot \vec{a}) + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} = |\vec{a}|^2 + |\vec{b}|^2$$

$$\Rightarrow \cancel{|\vec{a}|^2} + 2(\vec{a} \cdot \vec{b}) + \cancel{|\vec{b}|^2} = \cancel{|\vec{a}|^2} + \cancel{|\vec{b}|^2}$$

$$\Rightarrow 2(\vec{a} \cdot \vec{b}) = 0$$

$$\Rightarrow \boxed{\vec{a} \cdot \vec{b} = 0} \quad \begin{array}{c} \vec{b} \\ \uparrow \\ \perp \\ \vec{a} \end{array}$$

Part (II) Let $\vec{a} \perp \vec{b} \Rightarrow \boxed{\vec{a} \cdot \vec{b} = 0}$

$$\begin{aligned} (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) &= \vec{a} \cdot \vec{a} + \underbrace{\vec{a} \cdot \vec{b}}_0 + \underbrace{\vec{b} \cdot \vec{a}}_0 + \vec{b} \cdot \vec{b} \\ &= |\vec{a}|^2 + 0 + 0 + |\vec{b}|^2 \end{aligned}$$

$$= |\vec{a}|^2 + |\vec{b}|^2$$

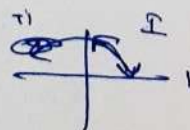
Q.16 If θ is the angle between two vectors $\vec{a}$ & $\vec{b}$, then $\vec{a} \cdot \vec{b} \geq 0$ only when — ~~(A) $0 < \theta < \frac{\pi}{2}$~~ ~~(B) $0 \leq \theta \leq \frac{\pi}{2}$~~ ~~(C) $0 < \theta < \pi$~~ ~~(D) $0 \leq \theta \leq \pi$~~

$$\Rightarrow |\vec{a}| |\vec{b}| \cos \theta \geq 0$$

$$\Rightarrow \boxed{\cos \theta \geq 0}$$

$$\cos \theta = 0 \quad \theta = \frac{\pi}{2}$$

$$\cos \theta > 0 \quad \text{in I-quadrant}$$



in I-quadrant

$$\theta \in \left[0, \frac{\pi}{2}\right)$$

Q.17 Let $\vec{a}$ & $\vec{b}$ be two unit vectors and θ is the angle between them. Then $\vec{a} + \vec{b}$ is a unit vector if

- (A) $\theta = \frac{\pi}{4}$ (B) $\theta = \frac{\pi}{3}$ (C) $\theta = \frac{\pi}{2}$ (D) $\theta = \frac{2\pi}{3}$

$\vec{a} \cdot \vec{a} = |\vec{a}|^2$

$|\vec{a}| = |\vec{b}| = 1, |\vec{a} + \vec{b}| = 1$

$\Rightarrow |\vec{a} + \vec{b}|^2 = 1$

$\Rightarrow (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = 1$

$\Rightarrow |\vec{a}|^2 + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + |\vec{b}|^2 = 1$

$\Rightarrow 1 + 2(\vec{a} \cdot \vec{b}) + 1 = 1$

$\Rightarrow \vec{a} \cdot \vec{b} = -\frac{1}{2}$
 $\Rightarrow |\vec{a}| |\vec{b}| \cos \theta = -\frac{1}{2}$
 $\Rightarrow \cos \theta = -\frac{1}{2}$
 $\theta = 120^\circ = \frac{2\pi}{3}$

Q.18. The value of $\hat{i} \cdot (\hat{j} \times \hat{k}) + \hat{j} \cdot (\hat{i} \times \hat{k}) + \hat{k} \cdot (\hat{i} \times \hat{j})$ is $\rightarrow$

- (A) 0 (B) -1
 (C) 1 (D) 3

$\hat{k} \times \hat{j} = -\hat{i}$
 $\hat{i} \times \hat{k} = -\hat{j}$
 $\hat{i} \times \hat{j} = \hat{k}$
 $\hat{i} \cdot (-\hat{i}) + \hat{j} \cdot (-\hat{j}) + \hat{k} \cdot (\hat{k})$
 $= -1 - 1 + 1 = -1$

Q.19 If θ is the angle between any two vectors $\vec{a}$ & $\vec{b}$, then $|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$, when θ is equal to -

- (A) 0 (B) $\frac{\pi}{4}$
 (C) $\frac{\pi}{2}$ (D) π

$|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$
 $\Rightarrow |\vec{a}| |\vec{b}| \cos \theta = |\vec{a}| |\vec{b}| \sin \theta$

$\Rightarrow \cos \theta = \sin \theta$

$\Rightarrow 1 = \sin \theta$

$\Rightarrow \frac{\sin \theta}{\cos \theta} = 1$

$\Rightarrow \tan \theta = 1$
 $\theta = 45^\circ = \frac{\pi}{4}$